Rapporto Analisi sollecitazione

File analizzato:	Base.ipt
Versione di Autodesk Inventor:	2026.1 (Build 301233020, 233B)
Data di creazione:	21/09/2026, 12:32
Autore studio:	Nuzzolese
Riepilogo:	

Analisi statica:1

Obiettivo generale e impostazioni:

Obiettivo di progetto	Punto singolo
Tipo studio	Analisi statica
Data ultima modifica	21/09/2026, 12:31
Stato modello	[Primario]
Rileva ed elimina le modalità corpo rigido	No

iProperties

Riepilogo

Autore	nuzzo

Progetto

Numero parte	Base
Progettista	nuzzo
Costo preventivato	0,00 €
Data creazione	20/09/2026

Stato

Stato progetto	In corso

Fisiche

Materiale	Generico
Densità	1 g/cm^3
Massa	0,076886 kg
Area	181,198 cm^2
Volume	76,886 cm^3
Baricentro	x=0 cm y=1,19609 cm z=-0,0170201 cm

Nota: i valori fisici potrebbero essere diversi da quelli utilizzati per FEA riportati di seguito.

Impostazioni mesh:

Dimensione media elemento (frazione del diametro modello)	0,1
Dimensione minima elemento (frazione della dimensione media)	0,2
Fattore di gradazione	1,5
Angolo di rotazione massimo	60 gr
Crea elementi mesh curvi	Sì

Materiale/i

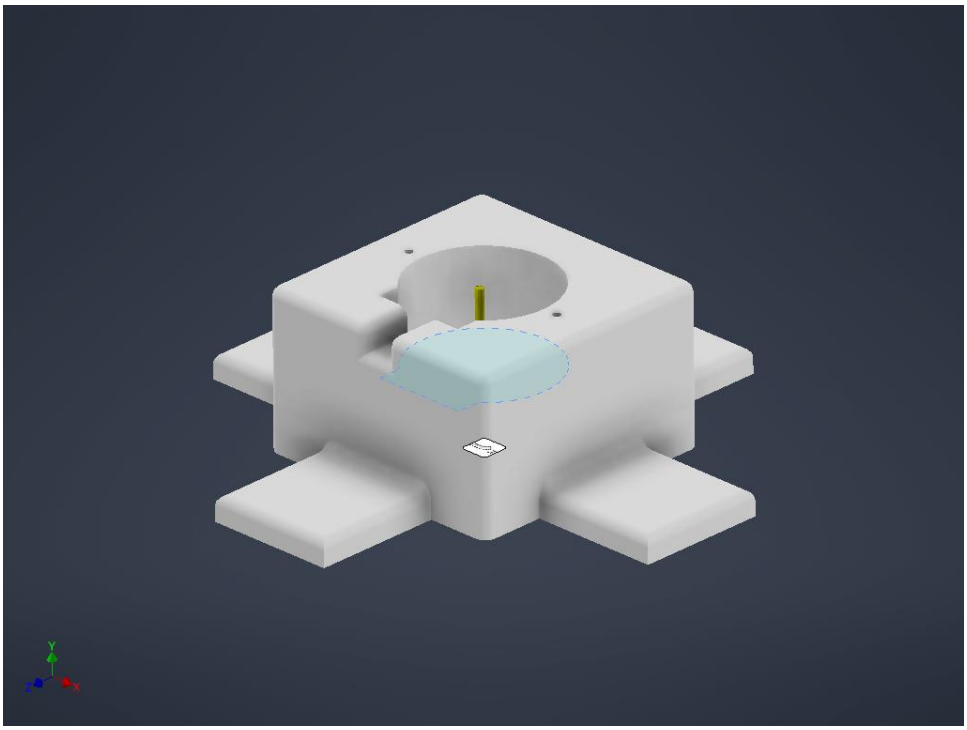
Nome	Plastica ABS	
Generale	Densità della massa	1,06 g/cm^3
	Resistenza allo snervamento	20 MPa
	Resistenza massima a trazione	29,6 MPa
Sollecitazione	Modulo di Young	2,24 GPa
	Coefficiente di Poisson	0,38 su
	Modulo a taglio	0,811594 GPa
Nome/i parte	Base.ipt	

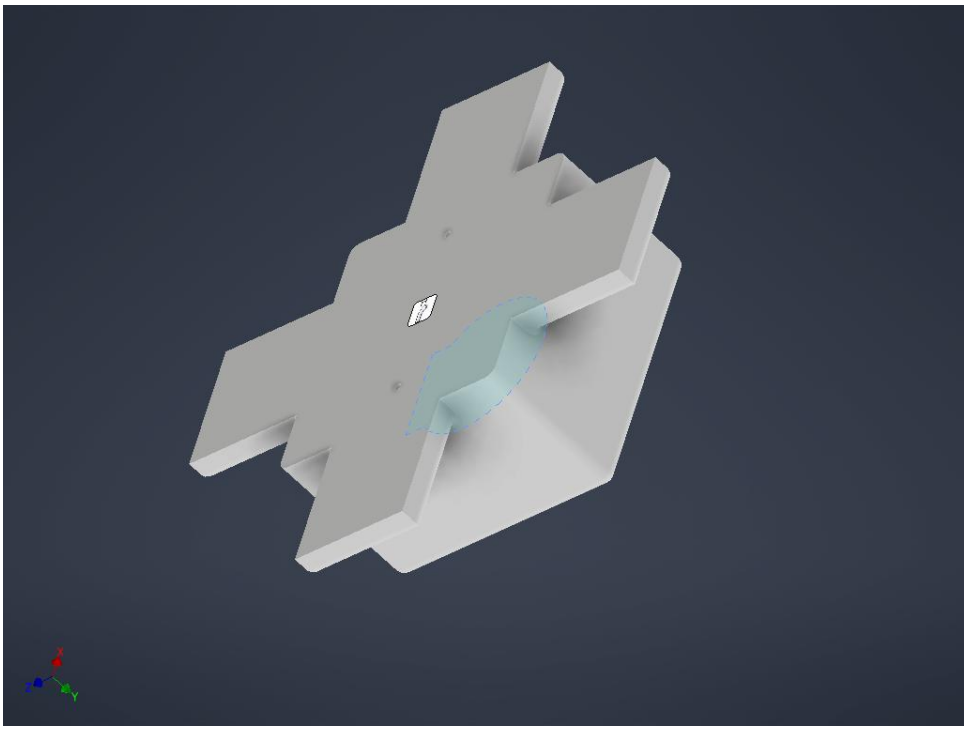
Condizioni operative

Forza:1

Tipo carico	Forza
Intensità	200,000 N
X vettore	0,000 N
Y vettore	-200,000 N
Z vettore	0,000 N

Faccia/e selezionate

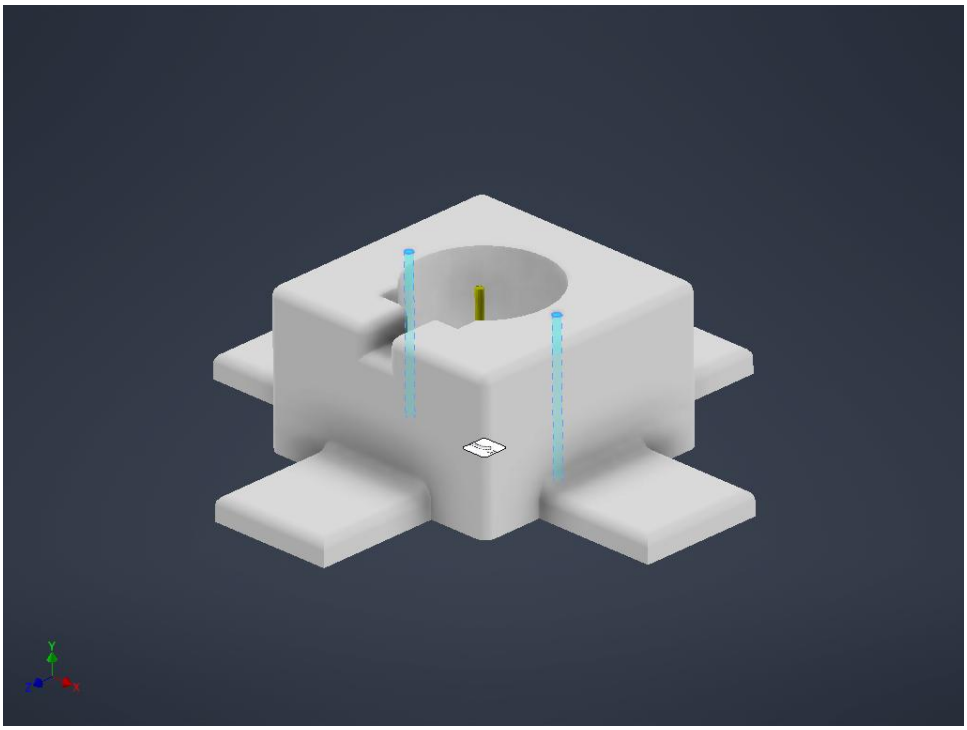


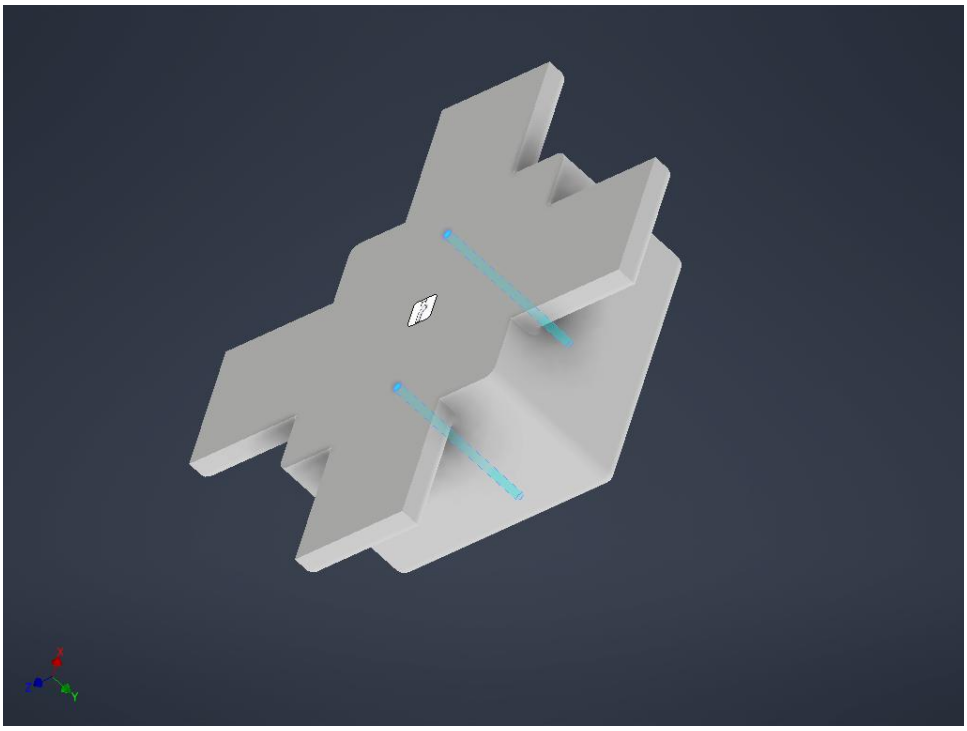


Momento:1

Tipo carico	Momento
Intensità	4000,000 N cm
X vettore	4000,000 N cm
Y vettore	0,000 N cm
Z vettore	0,000 N cm

Faccia/e selezionate

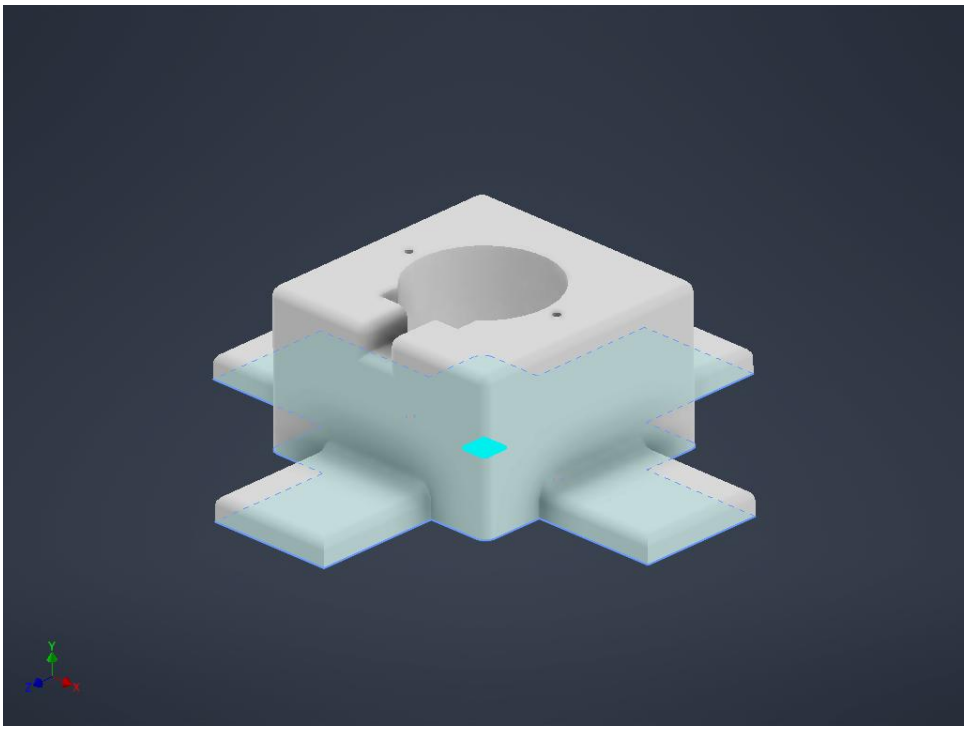


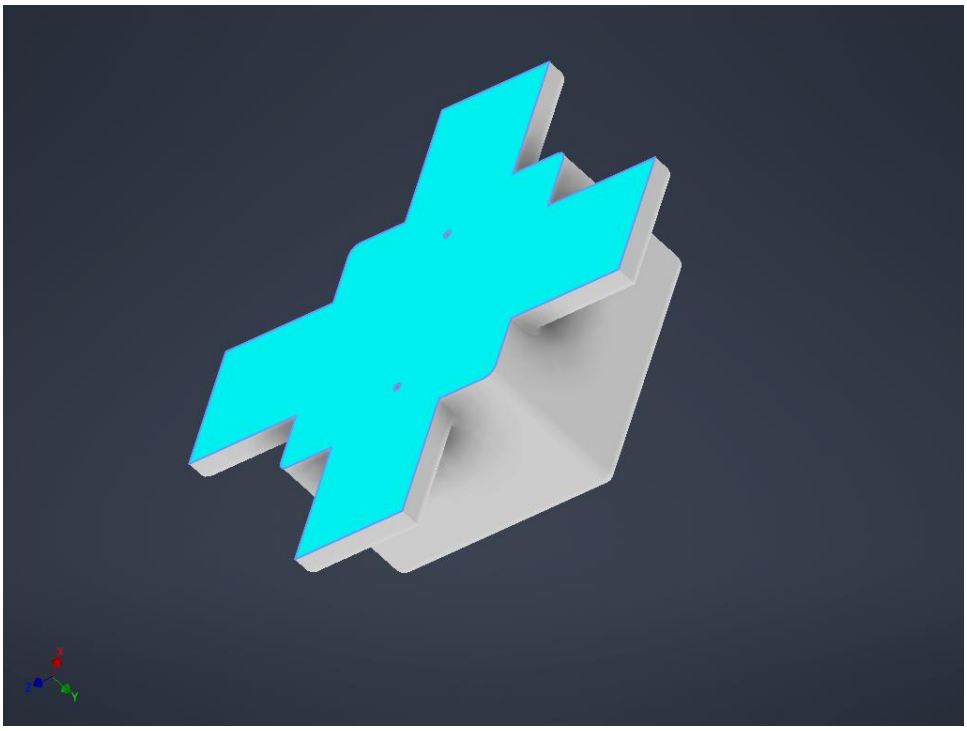


Vincolo di fissaggio:1

Tipo vincolo	Vincolo di fissaggio

Faccia/e selezionate





Risultati

Forza di reazione e momento su vincoli

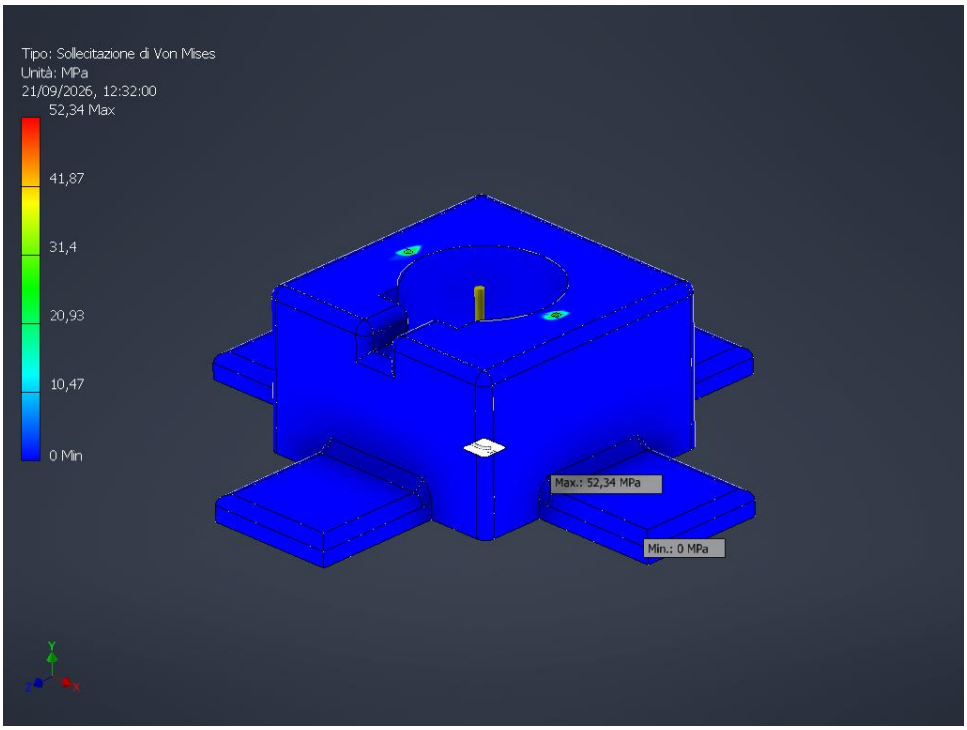
Nome vincolo	Forza di reazione		Momento di reazione	
	Intensità	Componente (X,Y,Z)	Intensità	Componente (X,Y,Z)
Vincolo di fissaggio:1	200,025 N	0 N	40,0928 N m	-40,0928 N m
		199,954 N		0 N m
		5,33614 N		0 N m

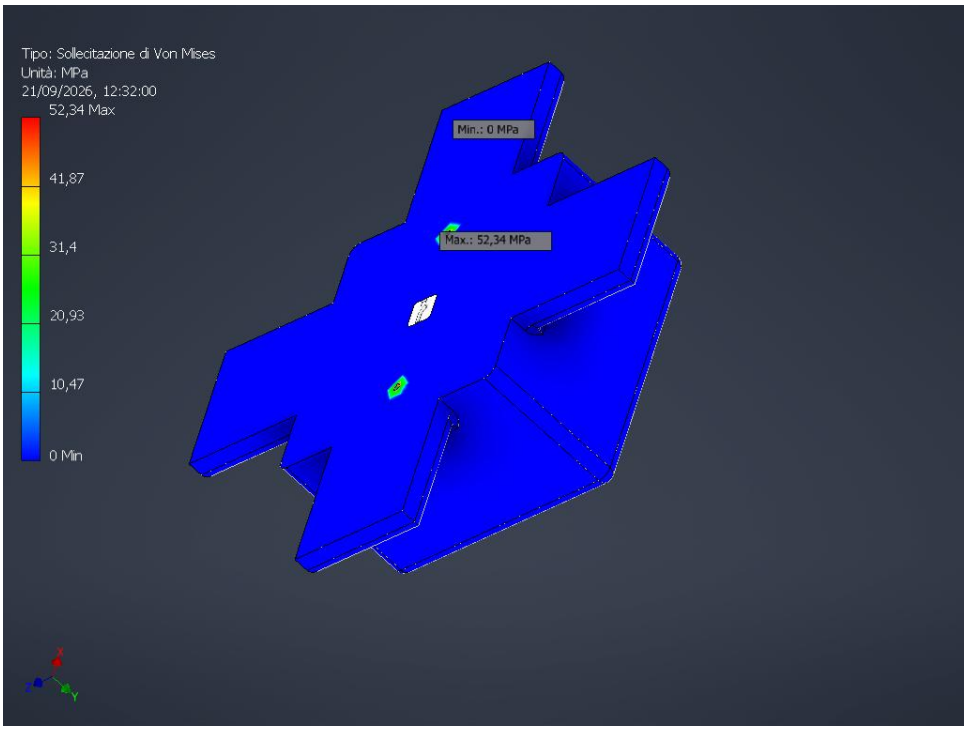
Riepilogo risultati

Nome	Minimo	Massimo
Volume	76886 mm^3	
Massa	0,0814992 kg	
Sollecitazione di Von Mises	0,0000478709 MPa	52,3365 MPa
Prima sollecitazione principale	-17,5507 MPa	43,7883 MPa
Terza sollecitazione principale	-40,1599 MPa	18,4738 MPa
Spostamento	0 mm	0,0782363 mm
Fattore di sicurezza	0,382143 su	15 su
Sollecitazione XX	-18,5199 MPa	19,6507 MPa
Sollecitazione XY	-6,18853 MPa	6,71547 MPa
Sollecitazione XZ	-23,5604 MPa	24,6251 MPa
Sollecitazione YY	-22,6197 MPa	20,3403 MPa
Sollecitazione YZ	-27,868 MPa	6,24929 MPa
Sollecitazione ZZ	-37,4155 MPa	35,073 MPa
Spostamento X	-0,0122928 mm	0,0124139 mm
Spostamento Y	-0,0196601 mm	0,0187655 mm
Spostamento Z	-0,0220296 mm	0,0778076 mm
Deformazione equivalente	0,000000019732 su	0,0215059 su
Prima deformazione principale	0,0000000148821 su	0,0197212 su
Terza deformazione principale	-0,0192704 su	-0,0000000116865 su
Deformazione XX	-0,00992829 su	0,0110616 su
Deformazione XY	-0,00381258 su	0,00413721 su
Deformazione XZ	-0,0145149 su	0,0151708 su
Deformazione YY	-0,00835228 su	0,00798558 su
Deformazione YZ	-0,0171687 su	0,00385001 su
Deformazione ZZ	-0,0143786 su	0,0141248 su

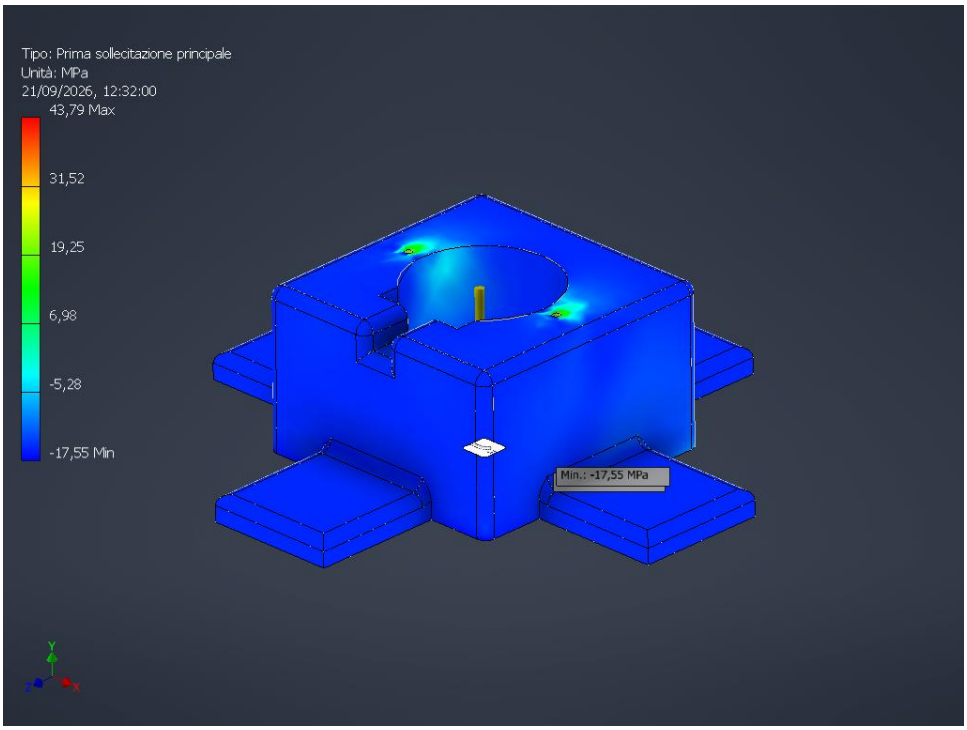
Figure

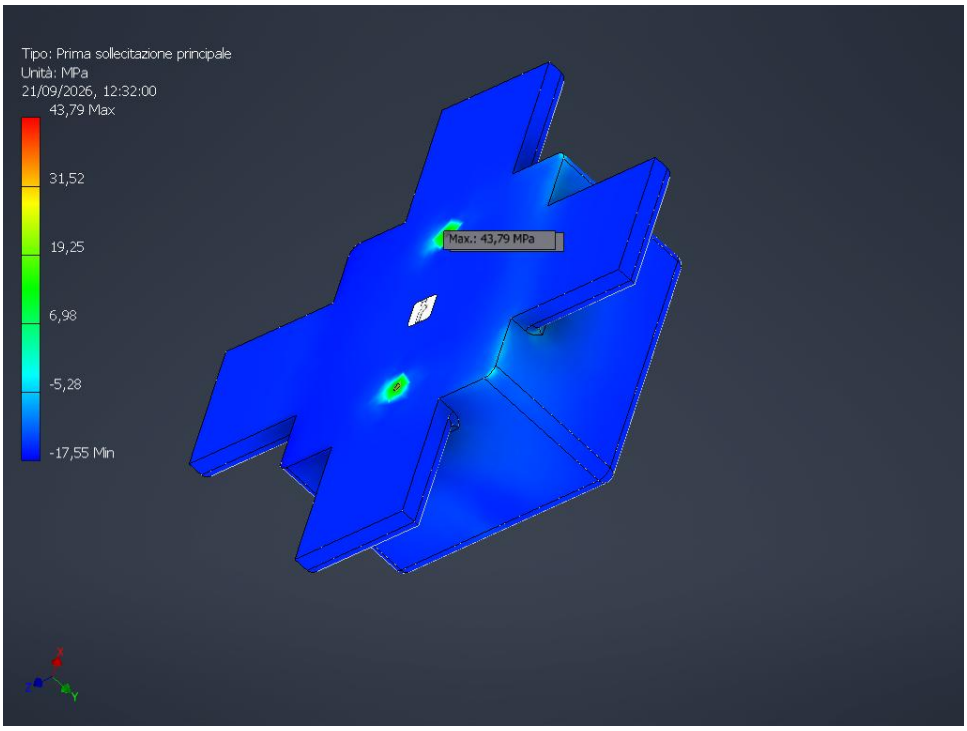
Sollecitazione di Von Mises



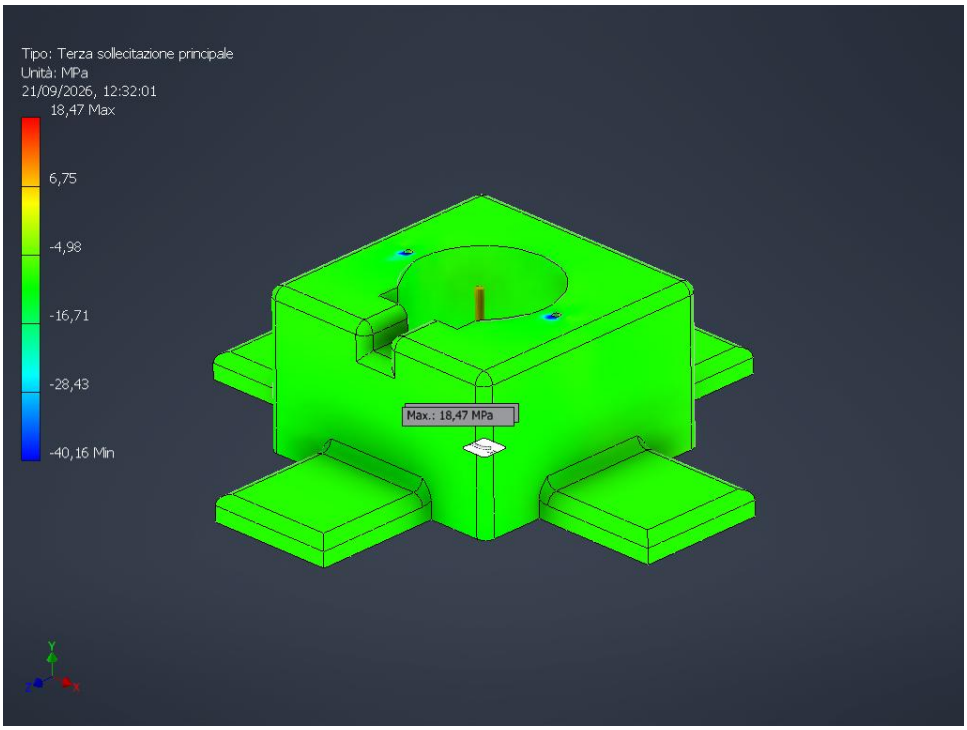


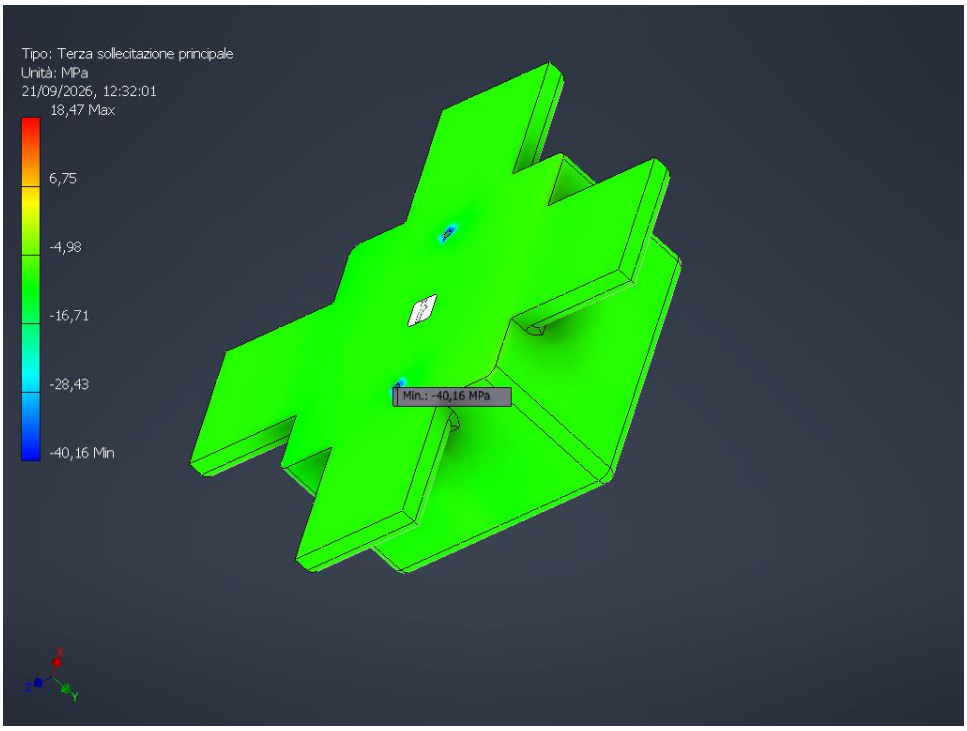
Prima sollecitazione principale



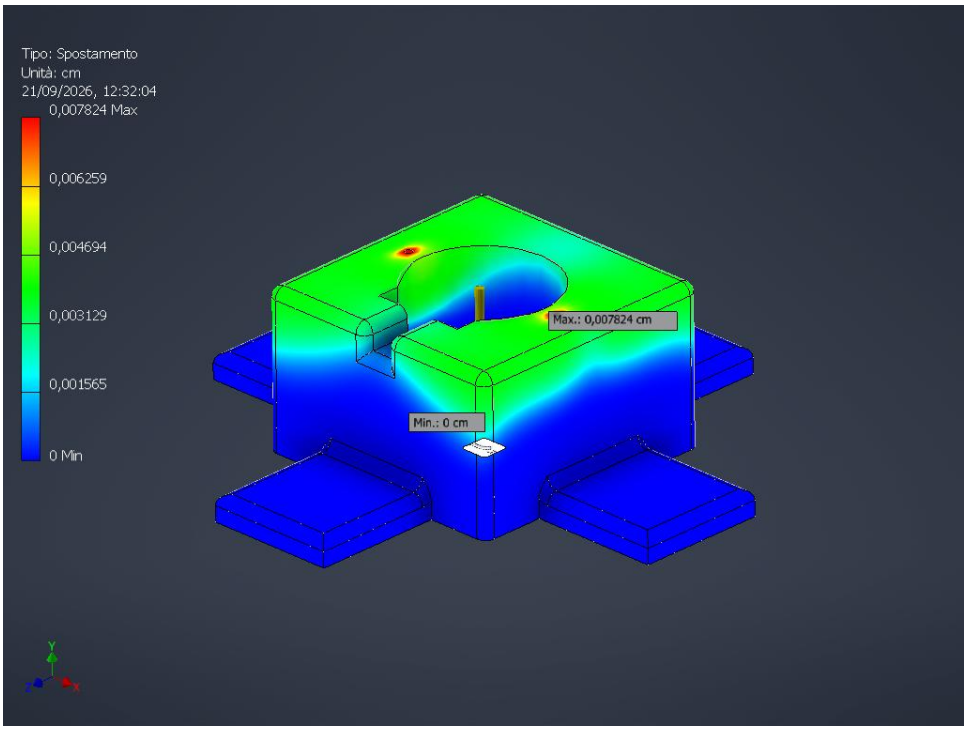


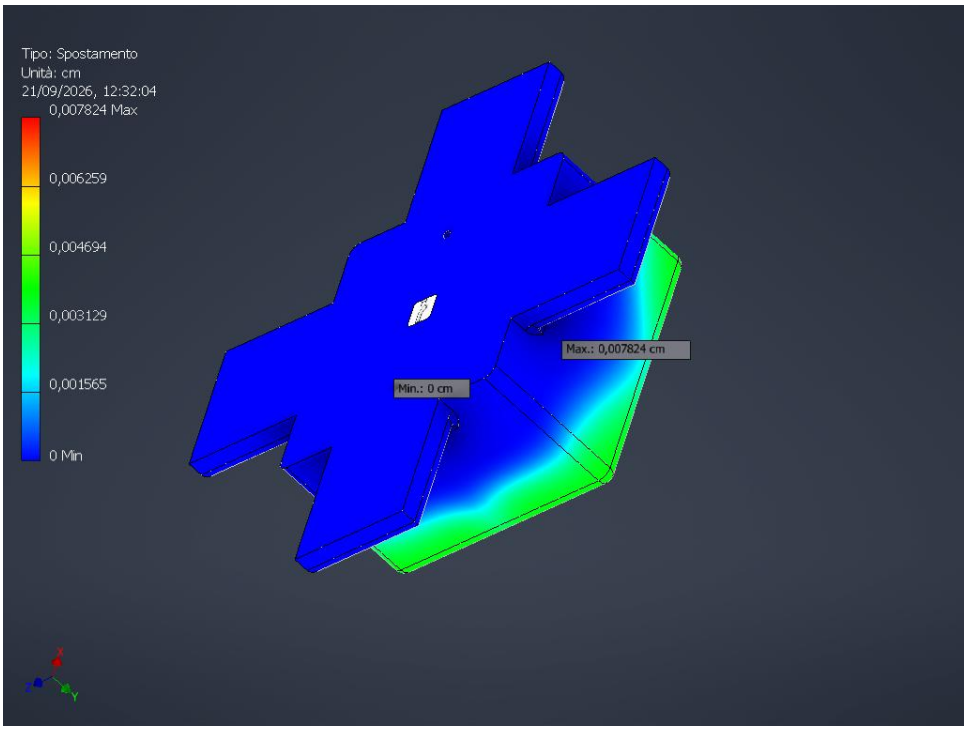
Terza sollecitazione principale



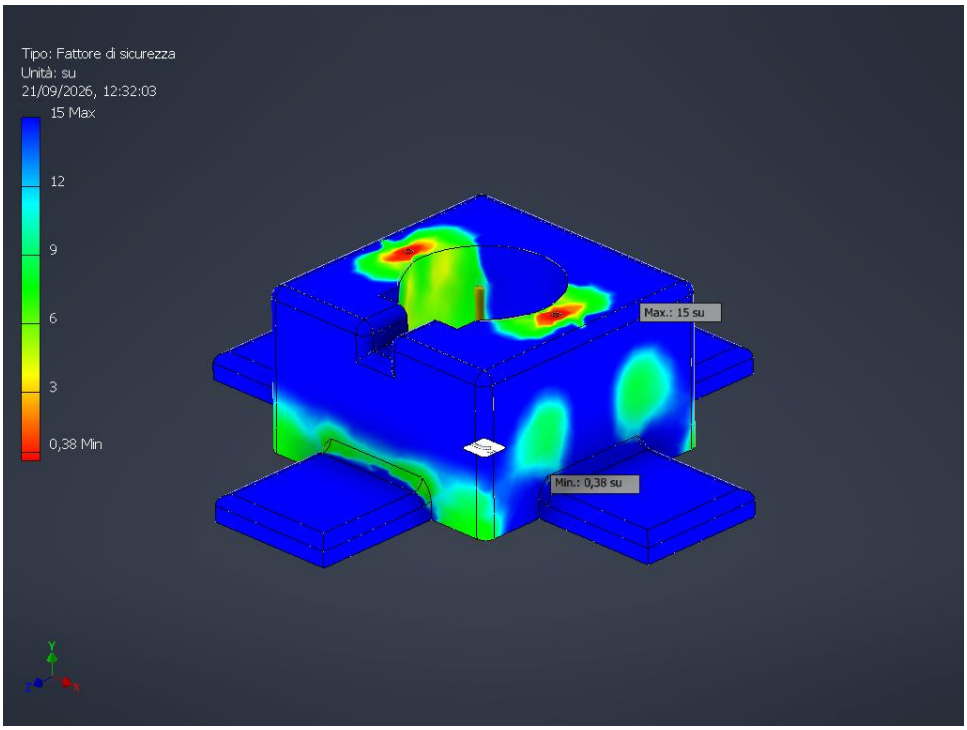


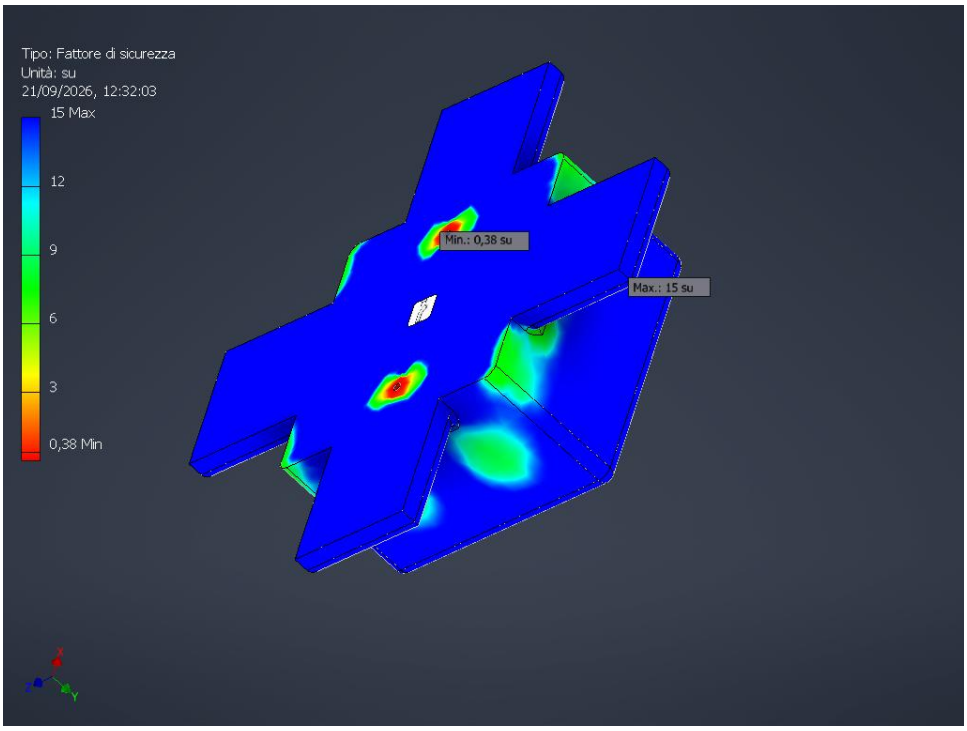
Spostamento



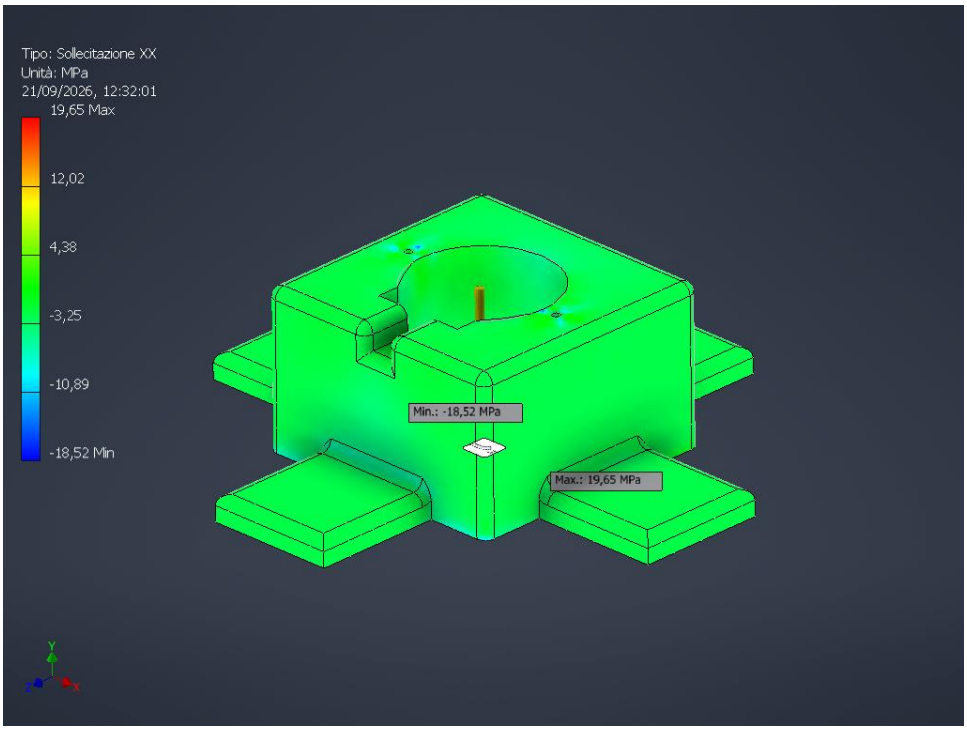


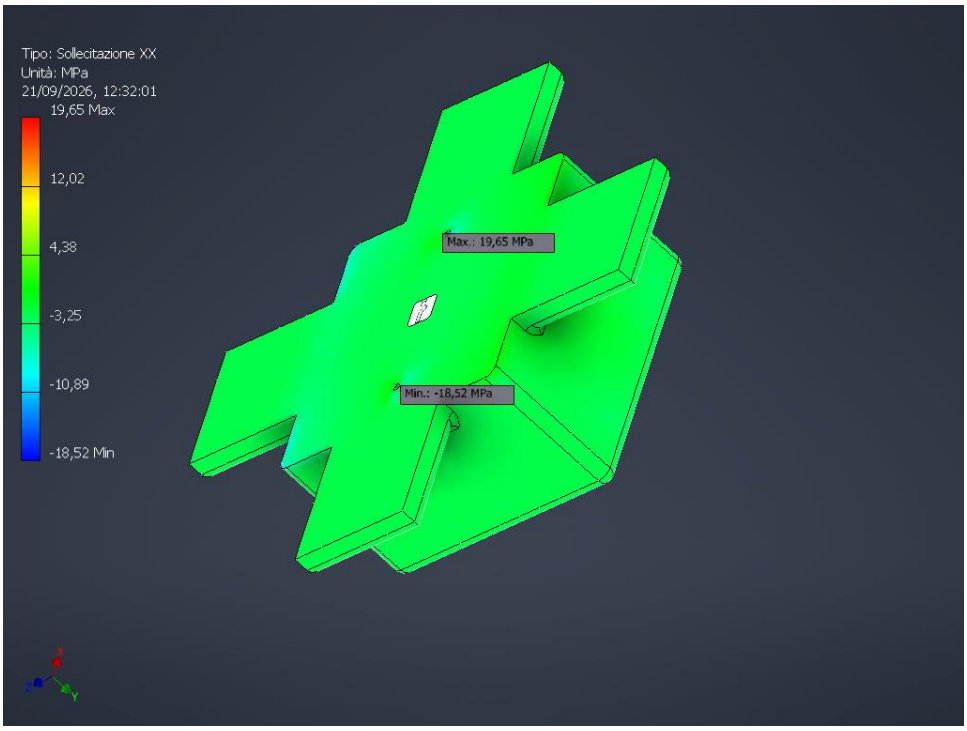
Fattore di sicurezza



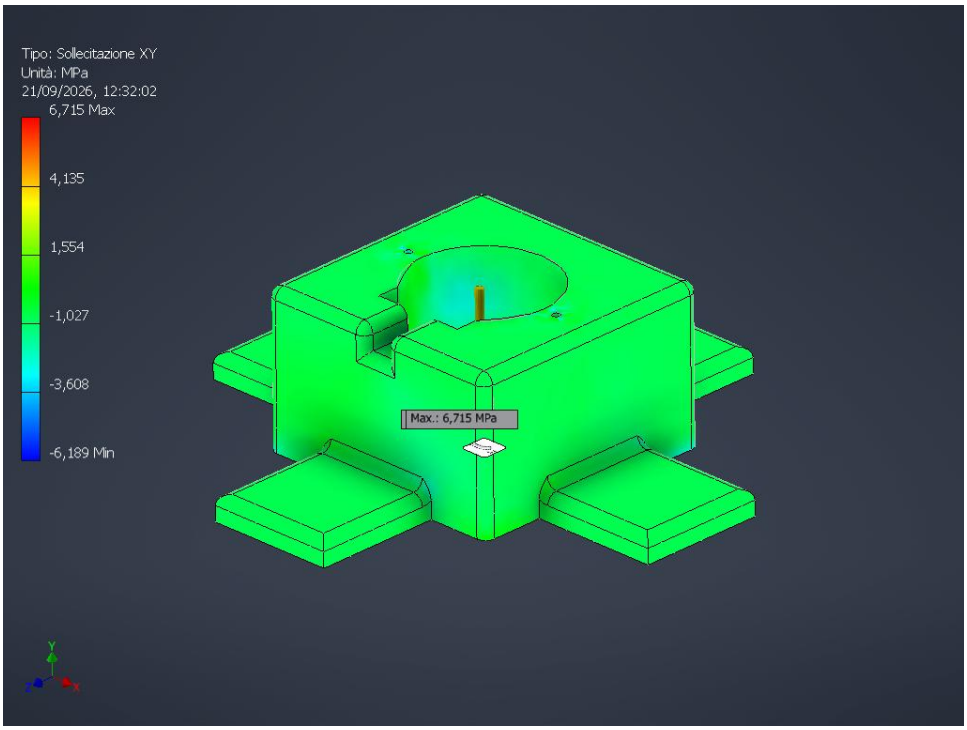


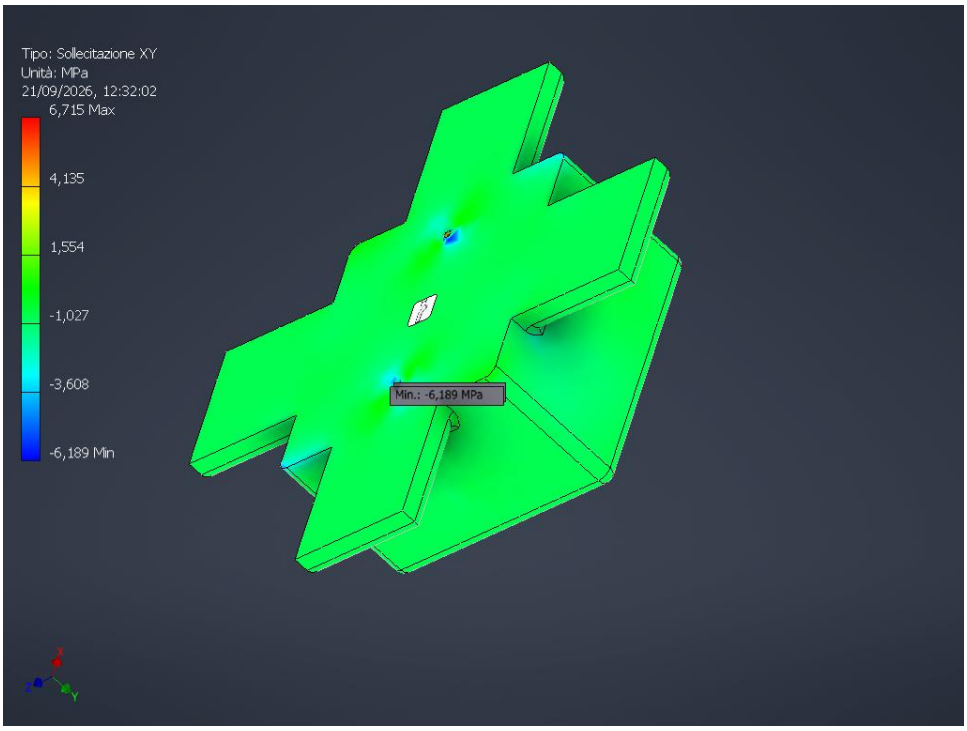
Sollecitazione XX



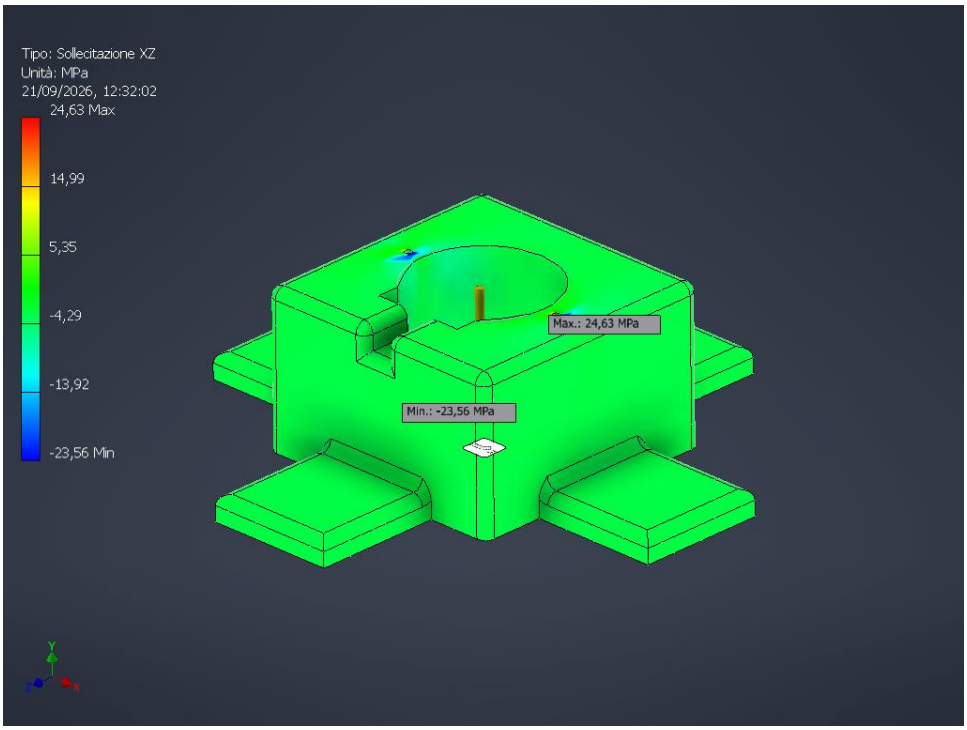


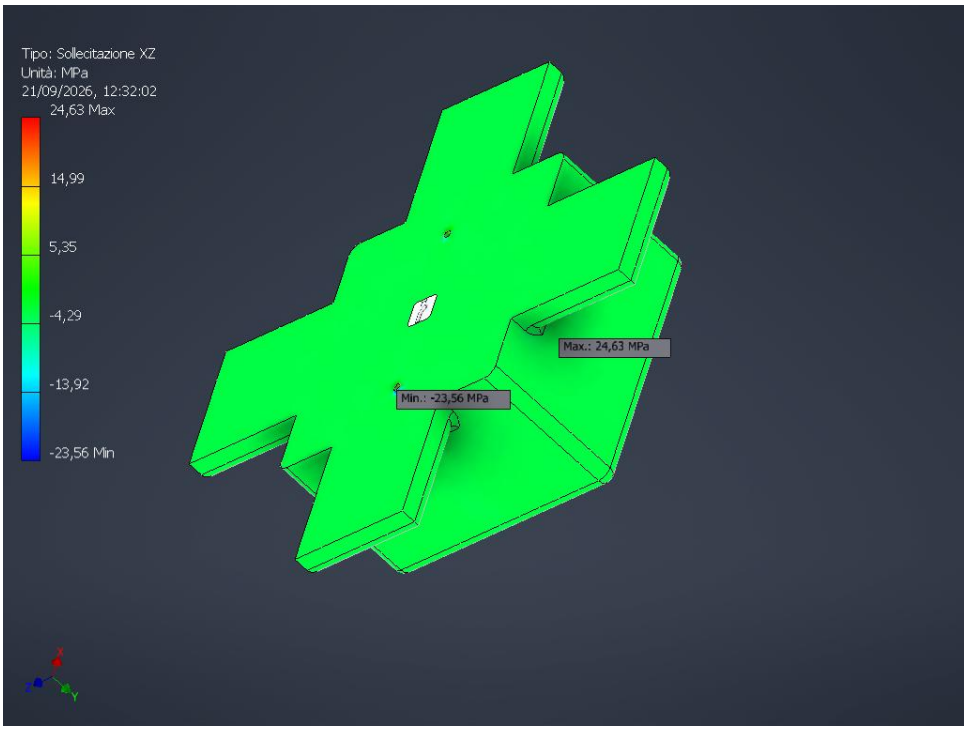
Sollecitazione XY



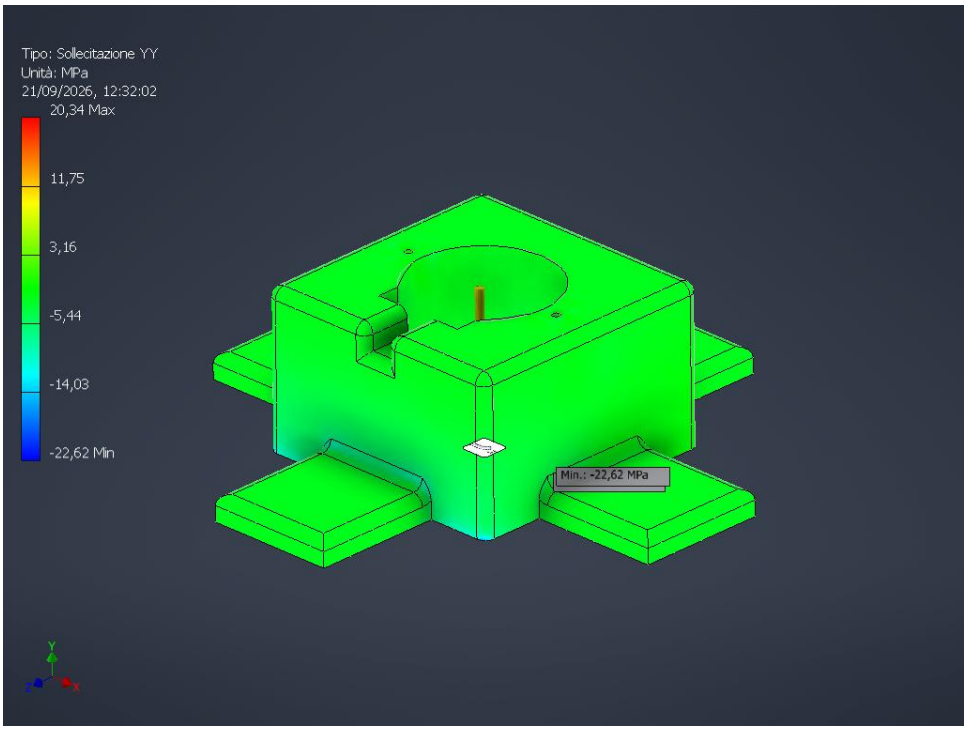


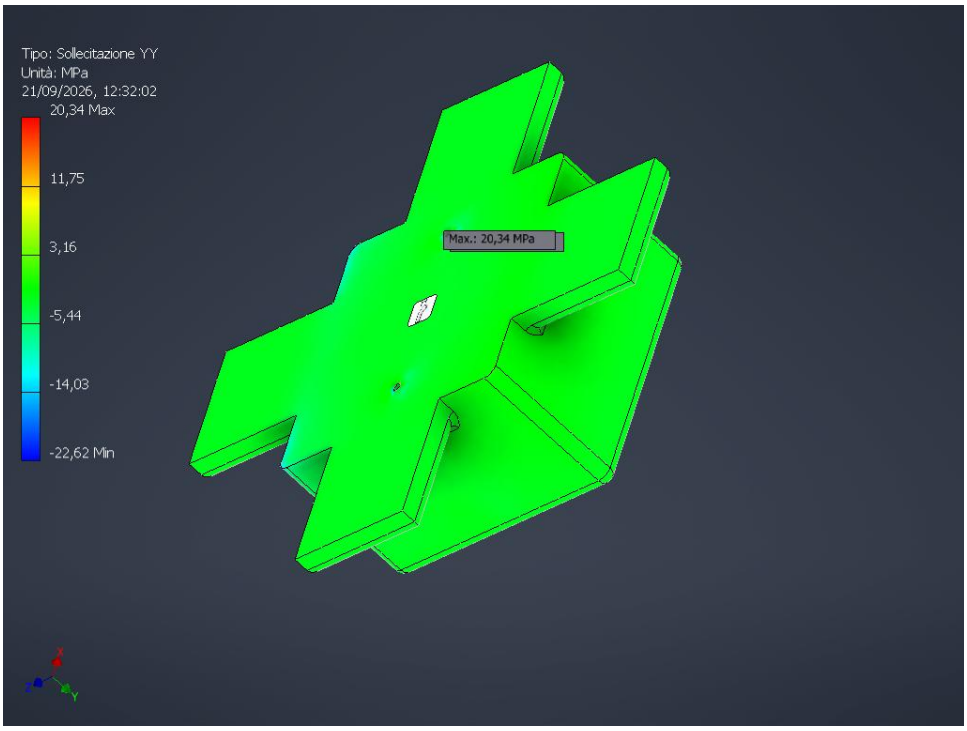
Sollecitazione XZ



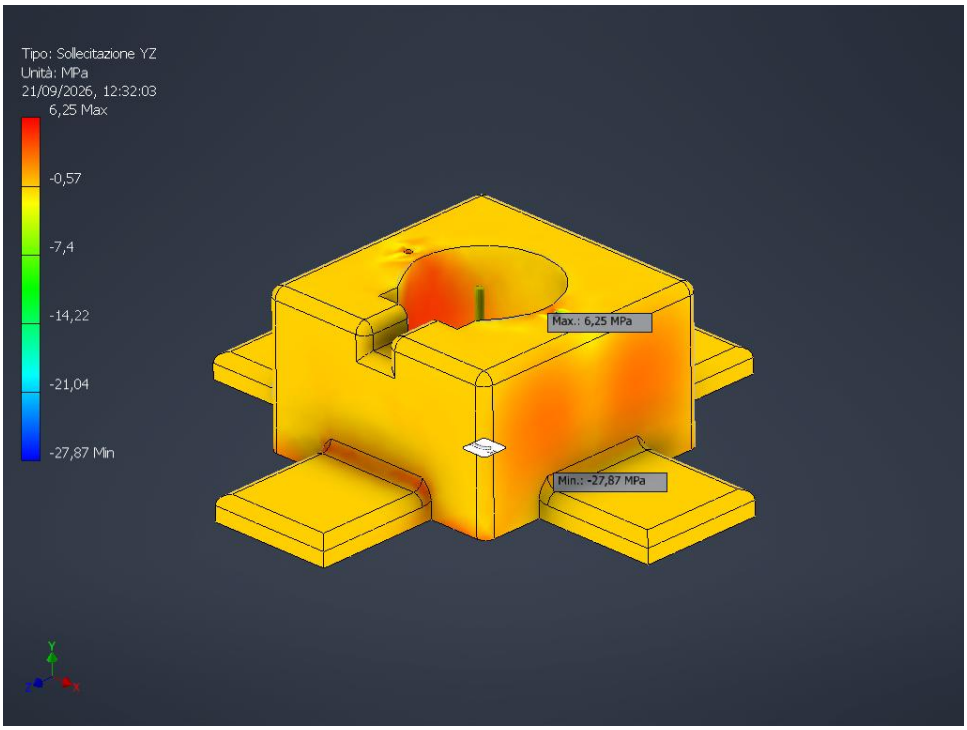


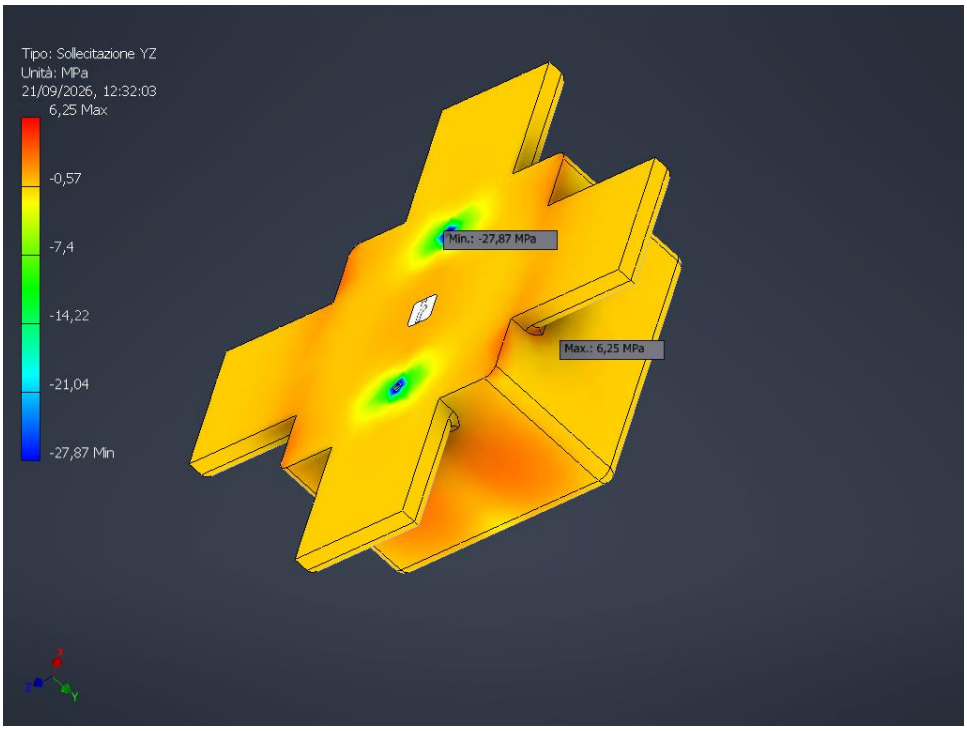
Sollecitazione YY



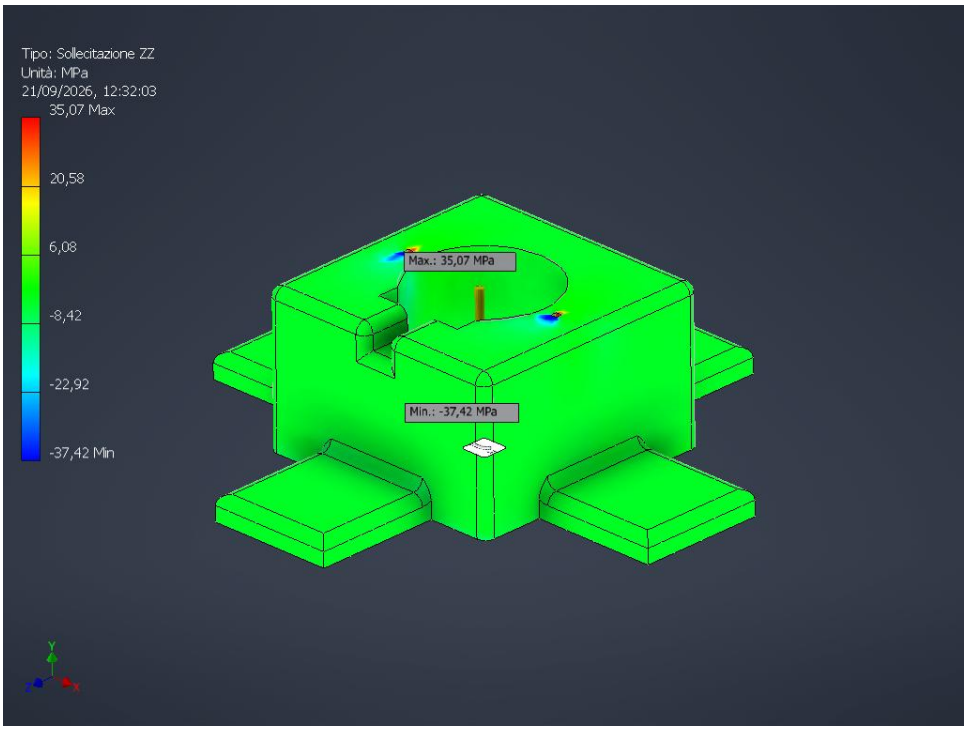


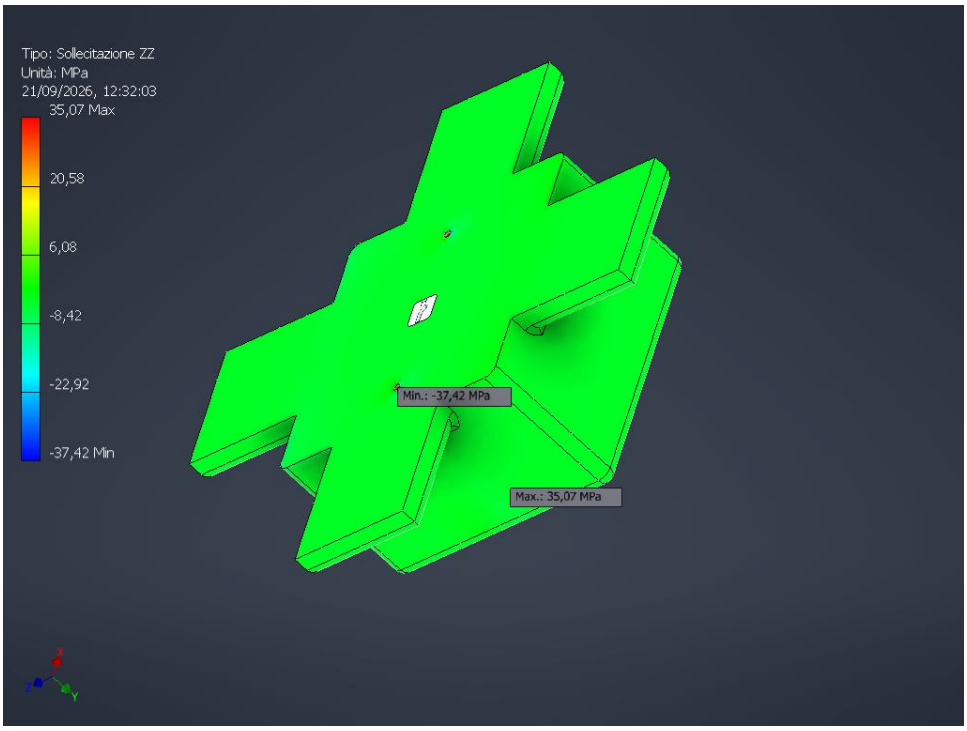
Sollecitazione YZ



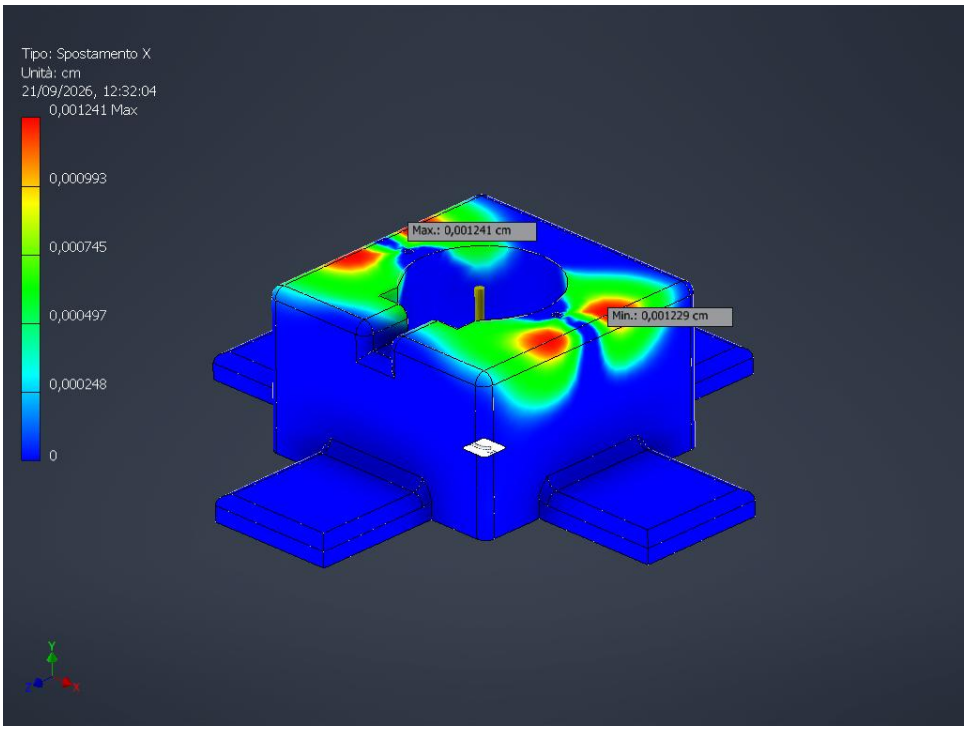


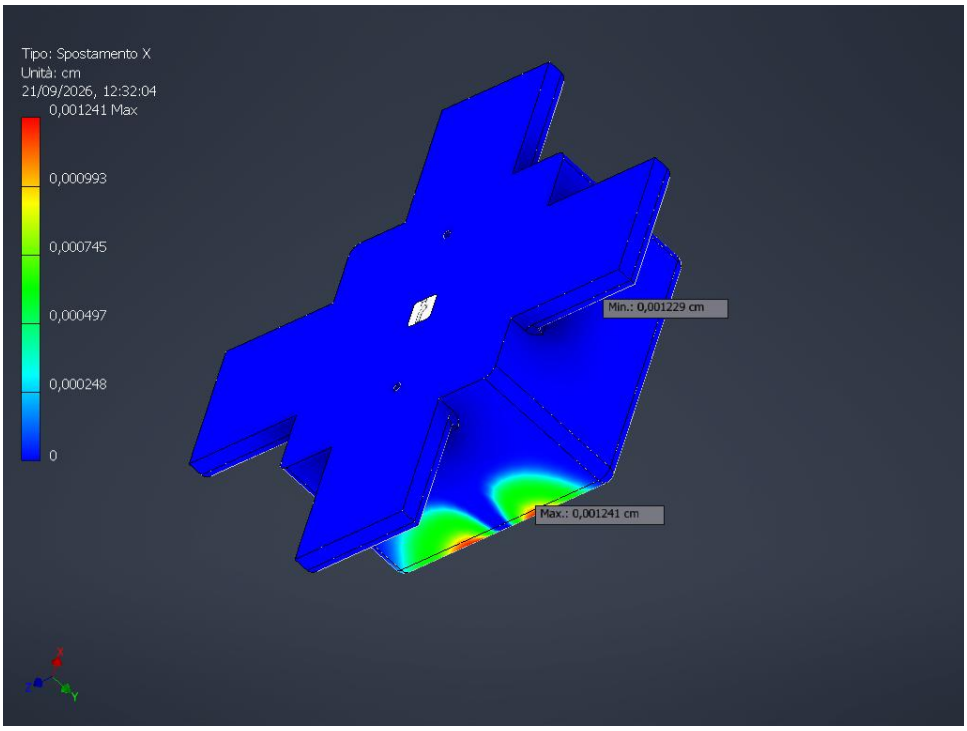
Sollecitazione ZZ



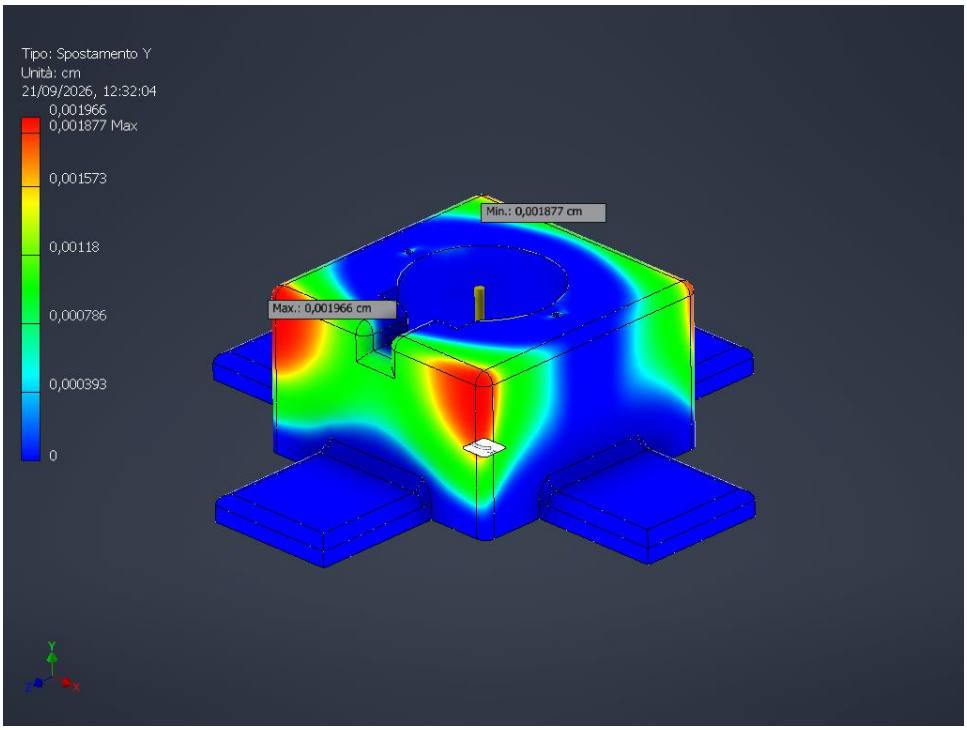


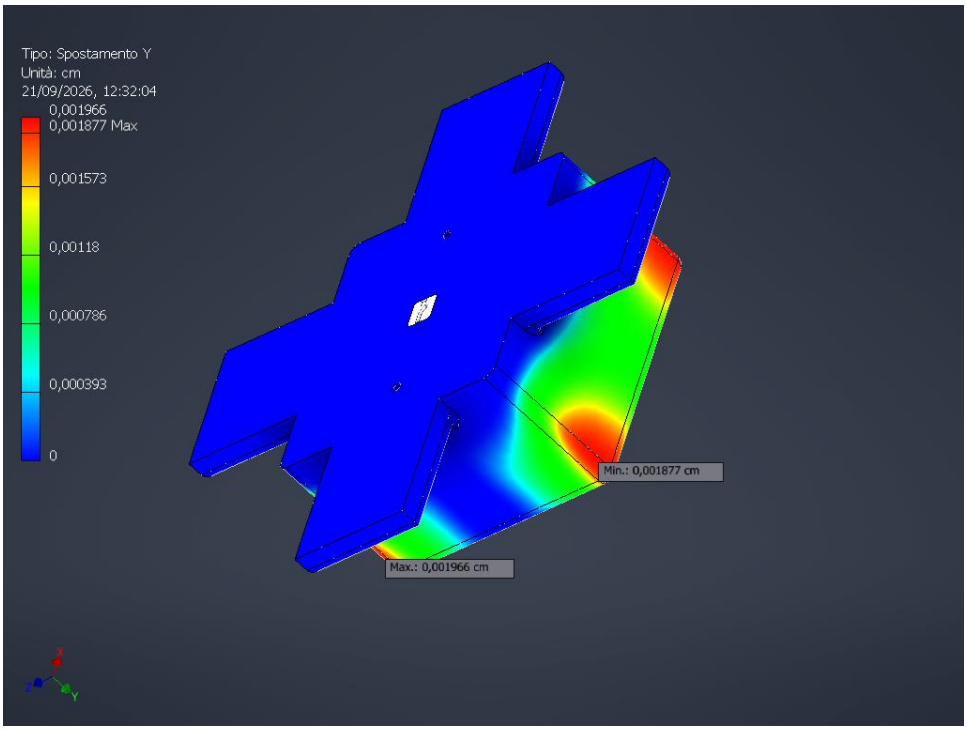
Spostamento X



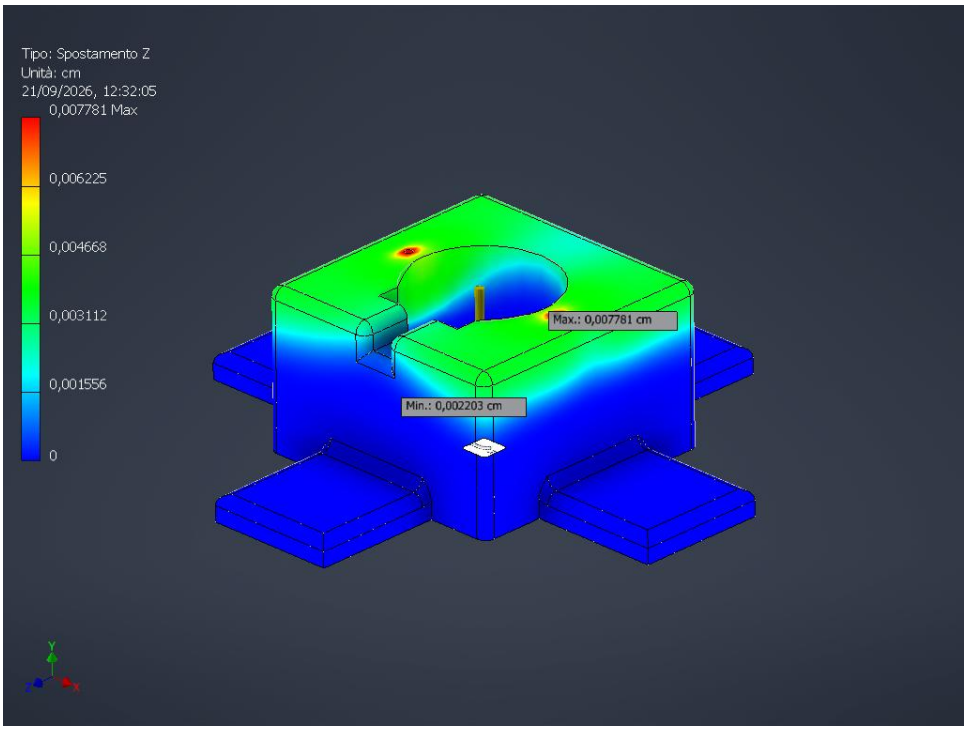


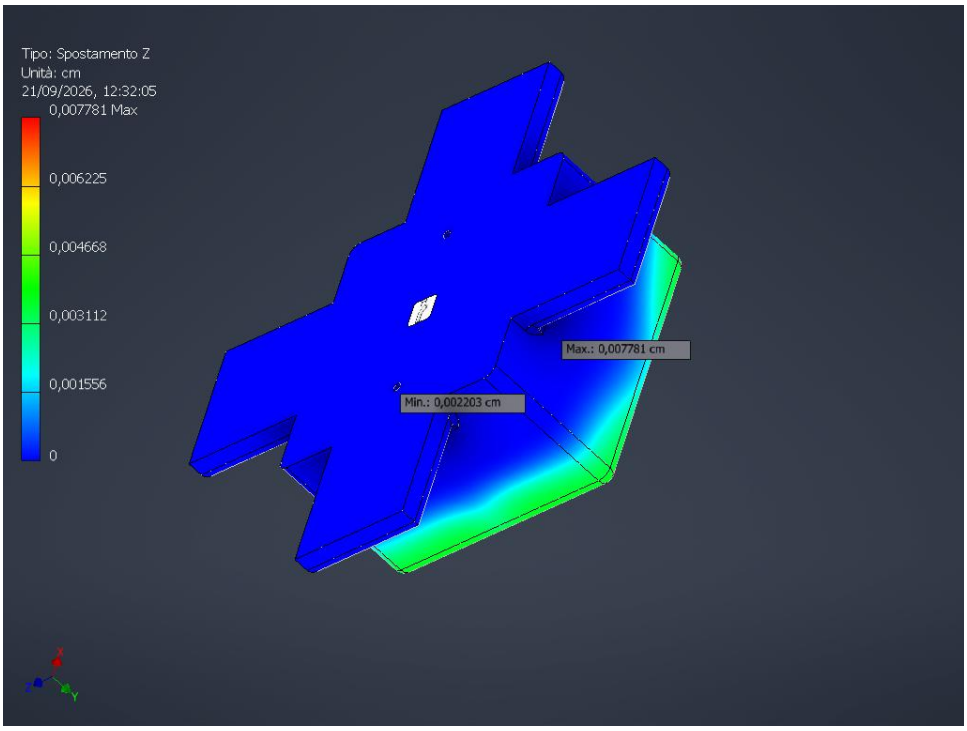
Spostamento Y



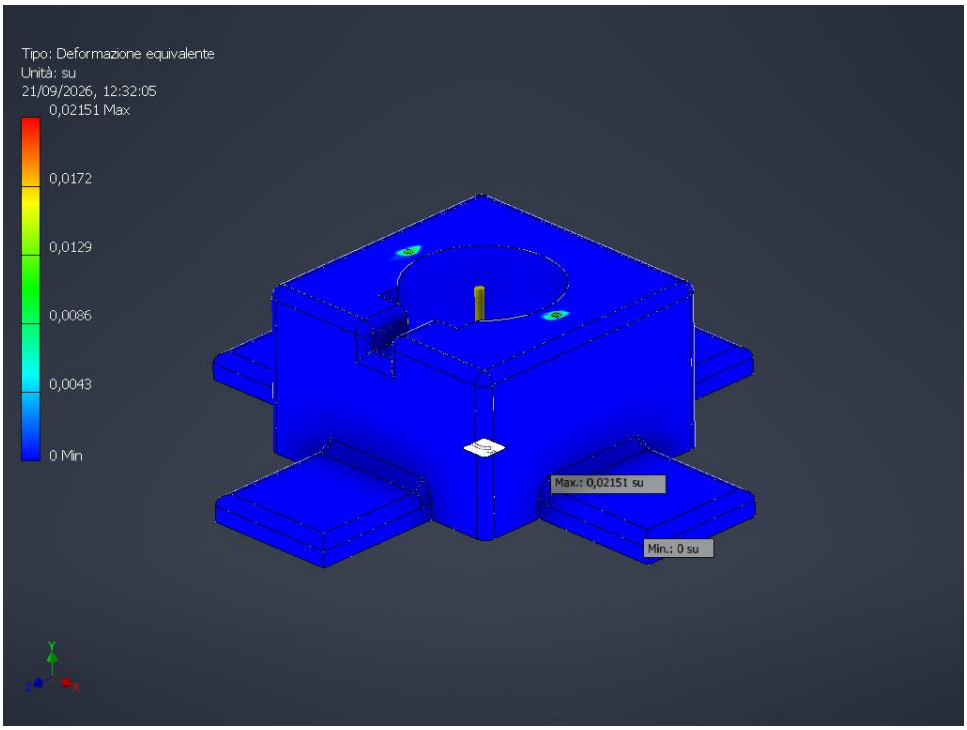


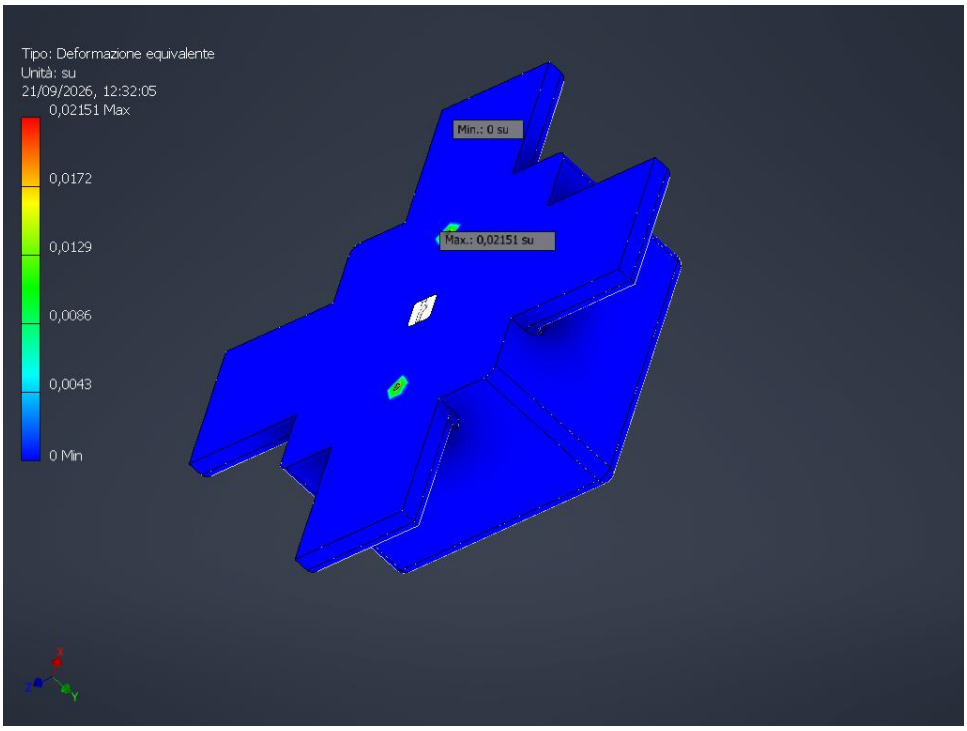
Spostamento Z



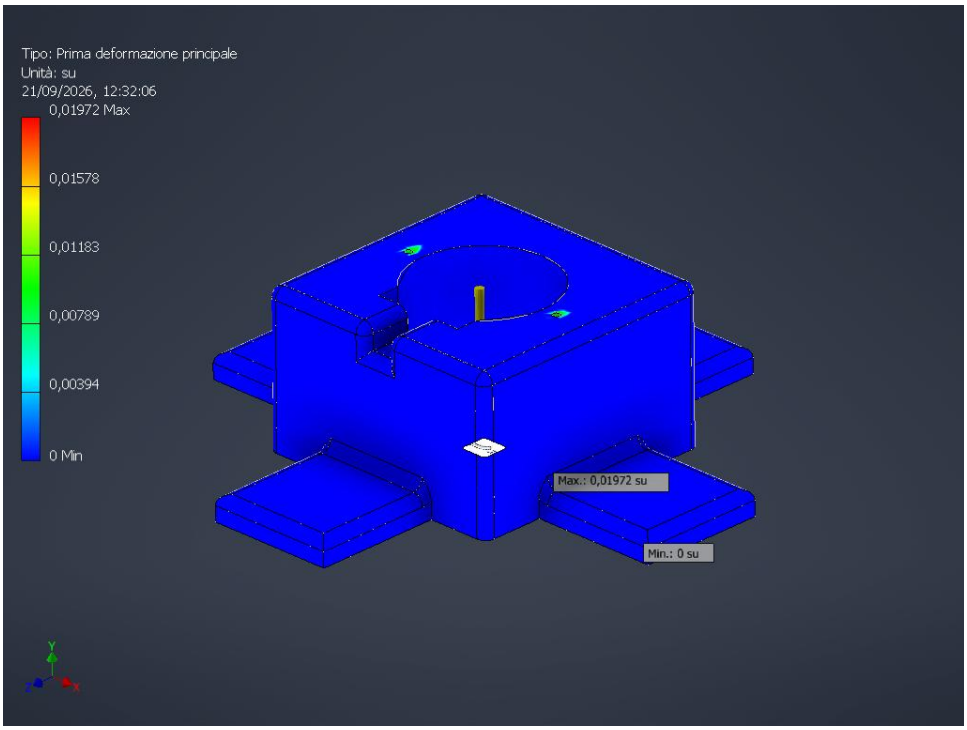


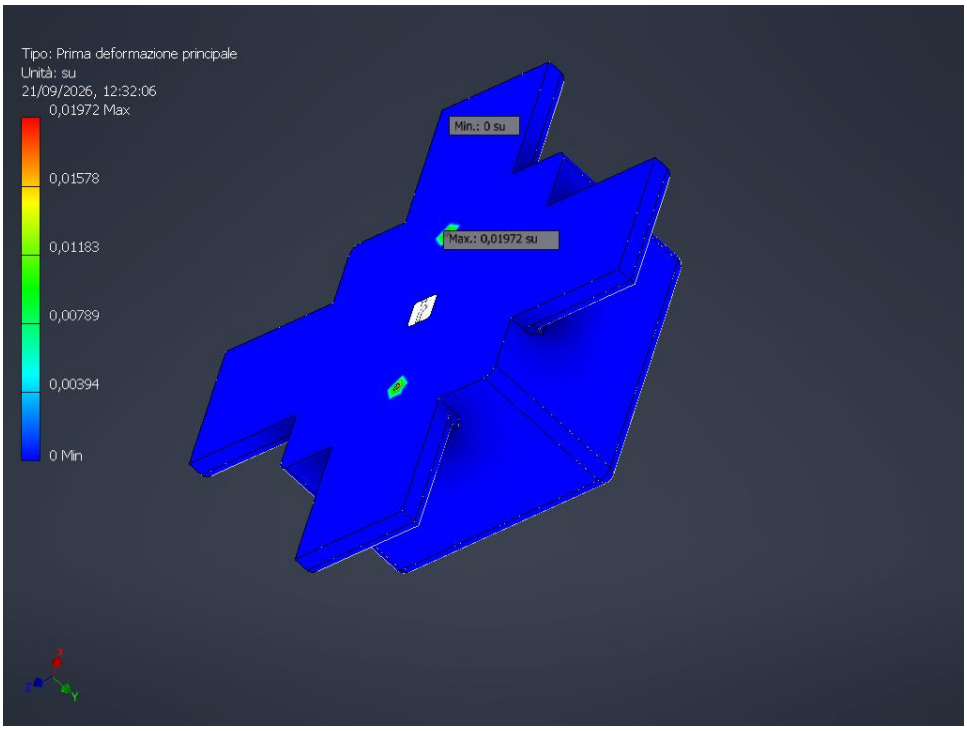
Deformazione equivalente



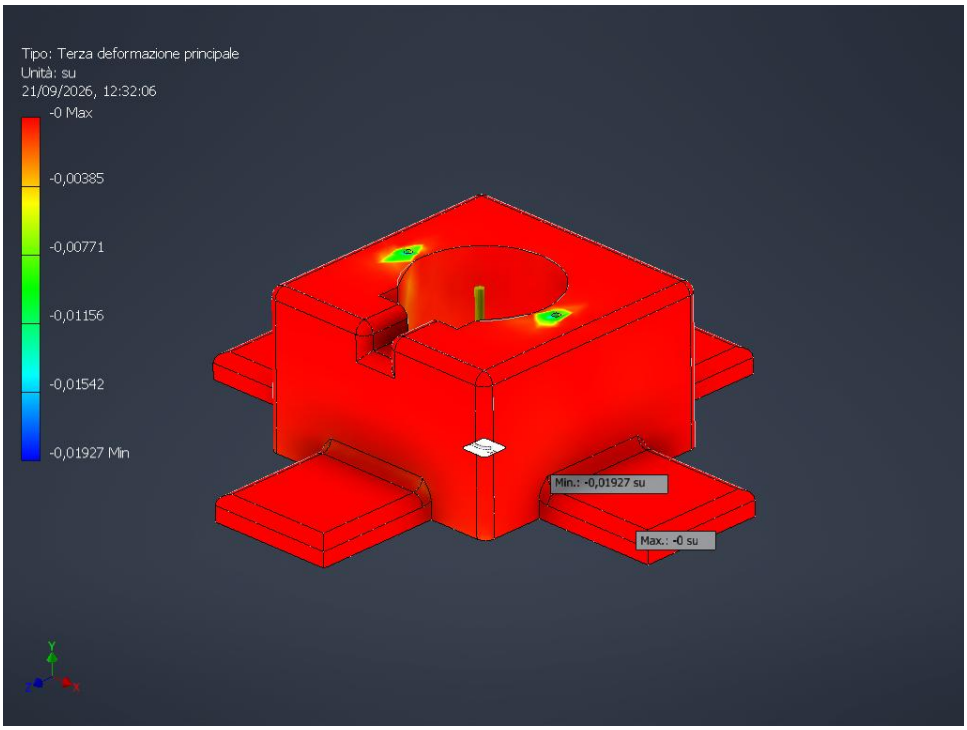


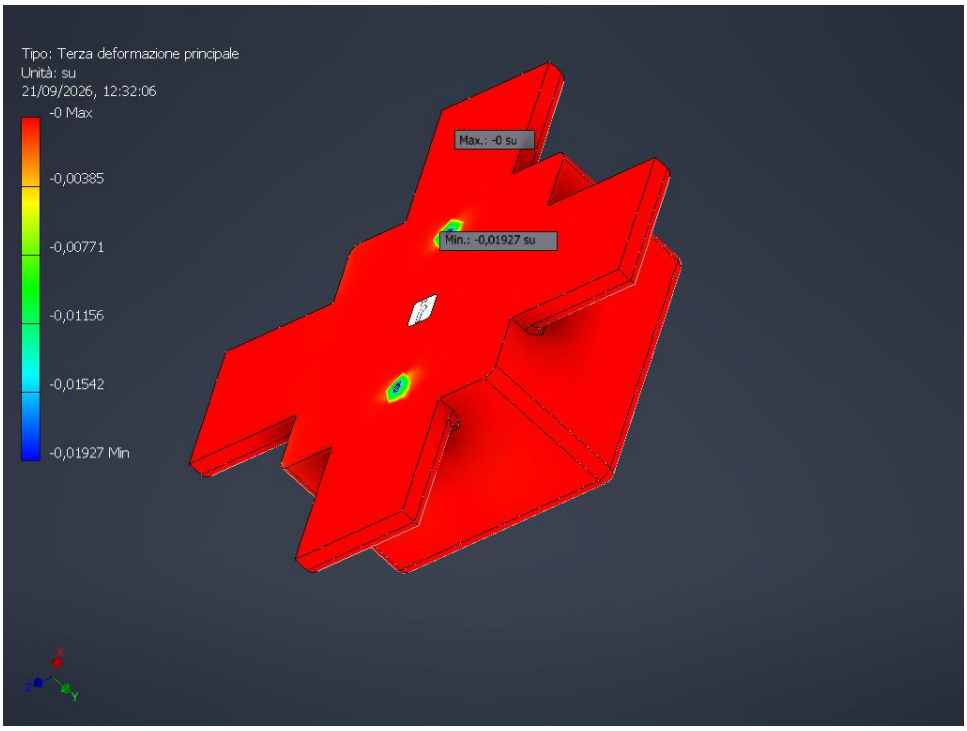
Prima deformazione principale



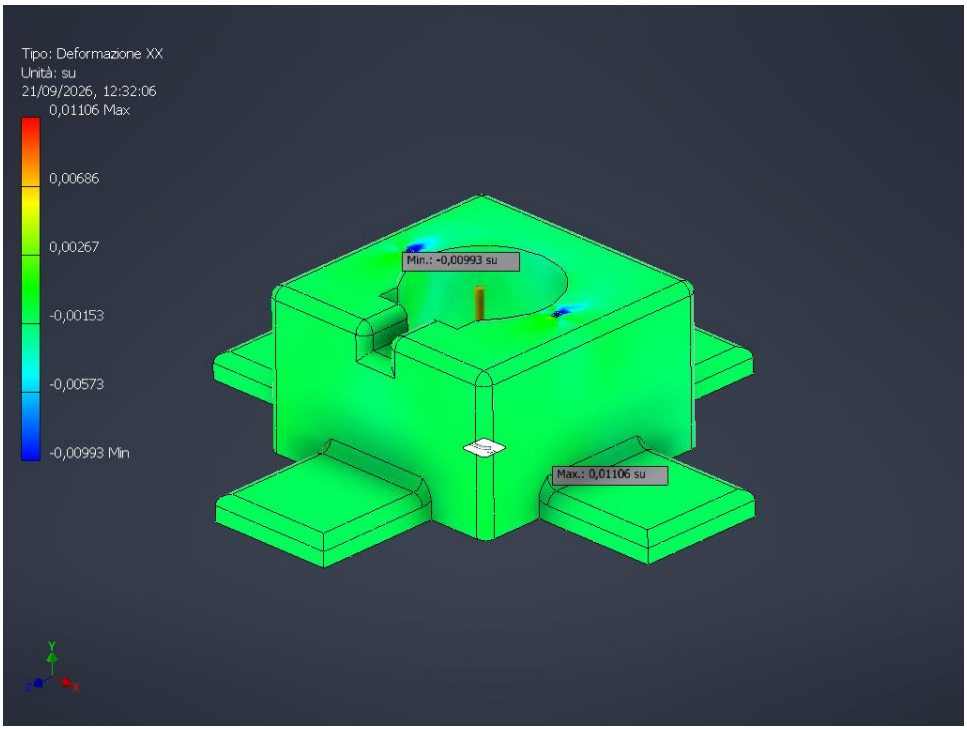


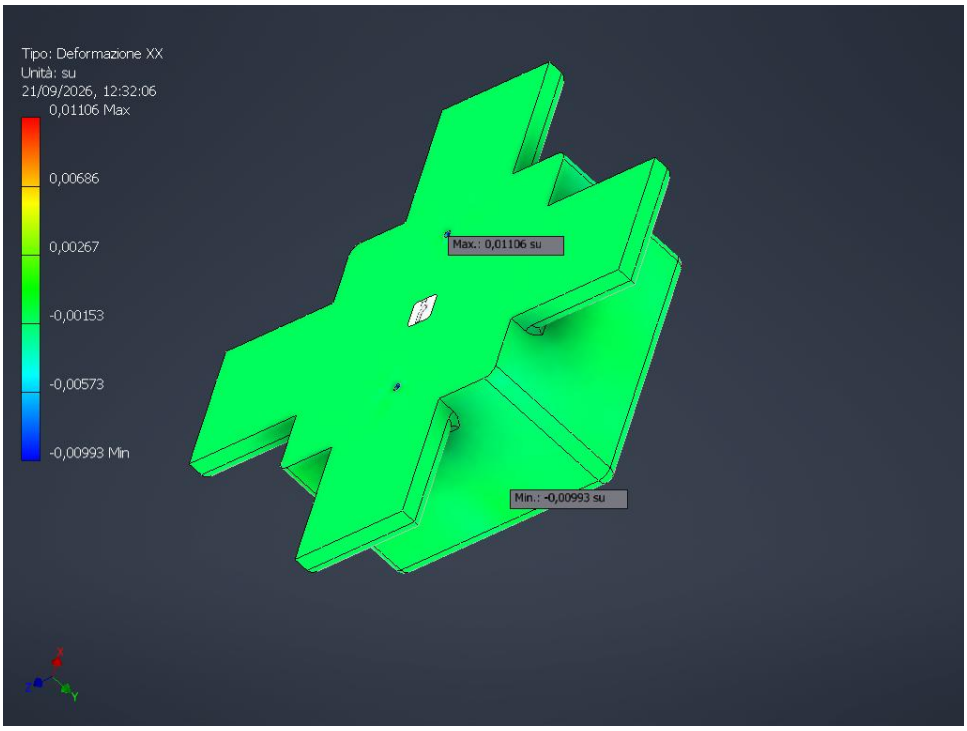
Terza deformazione principale



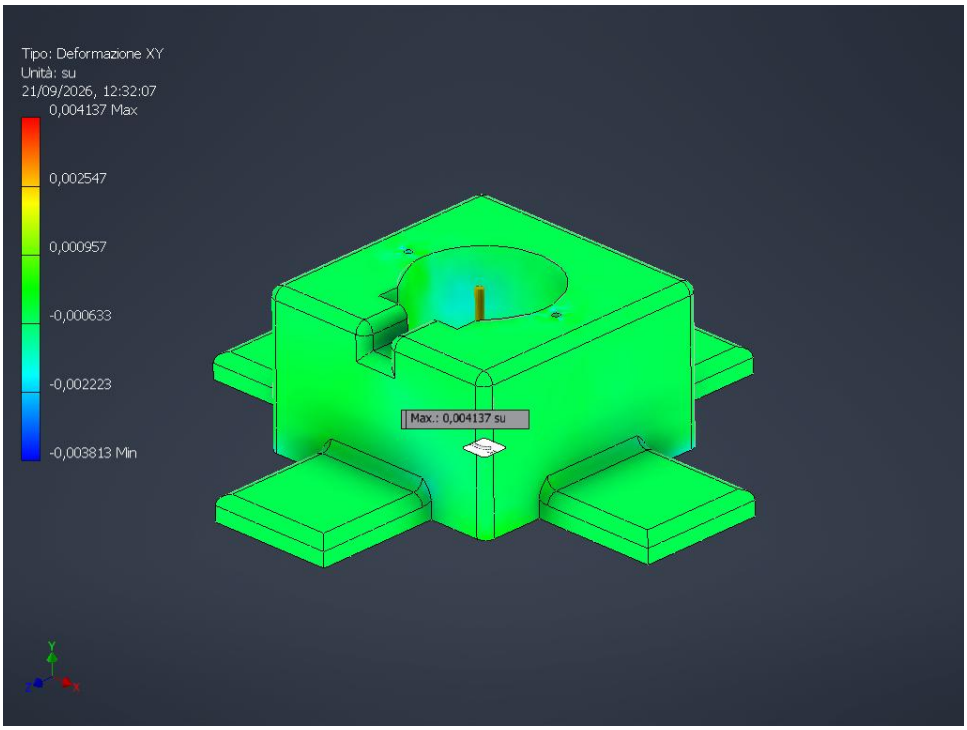


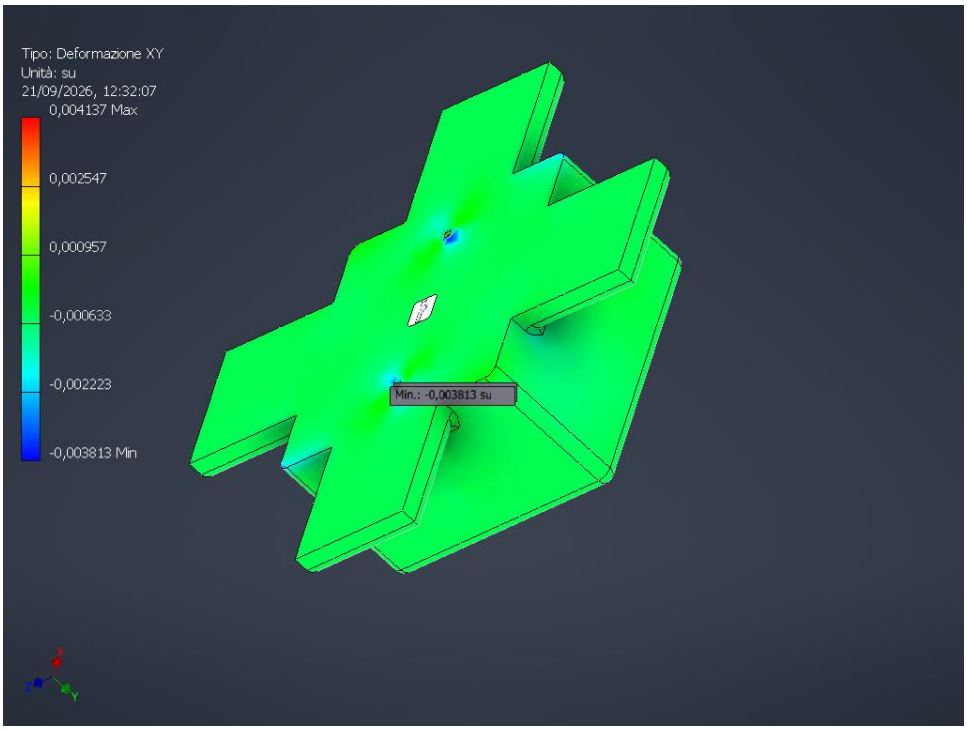
Deformazione XX



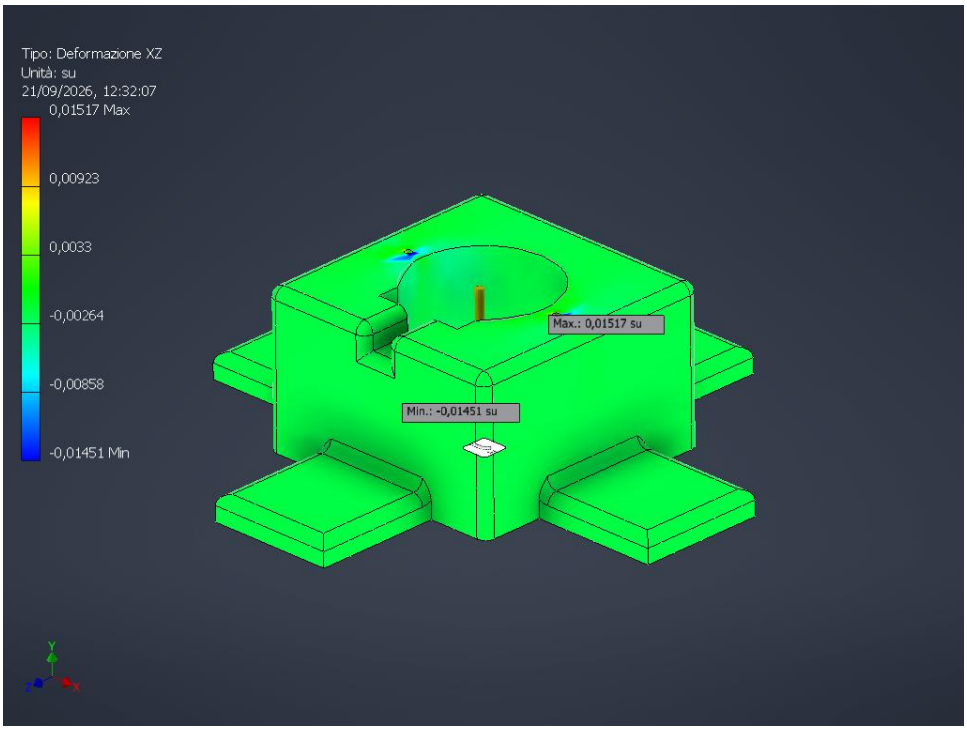


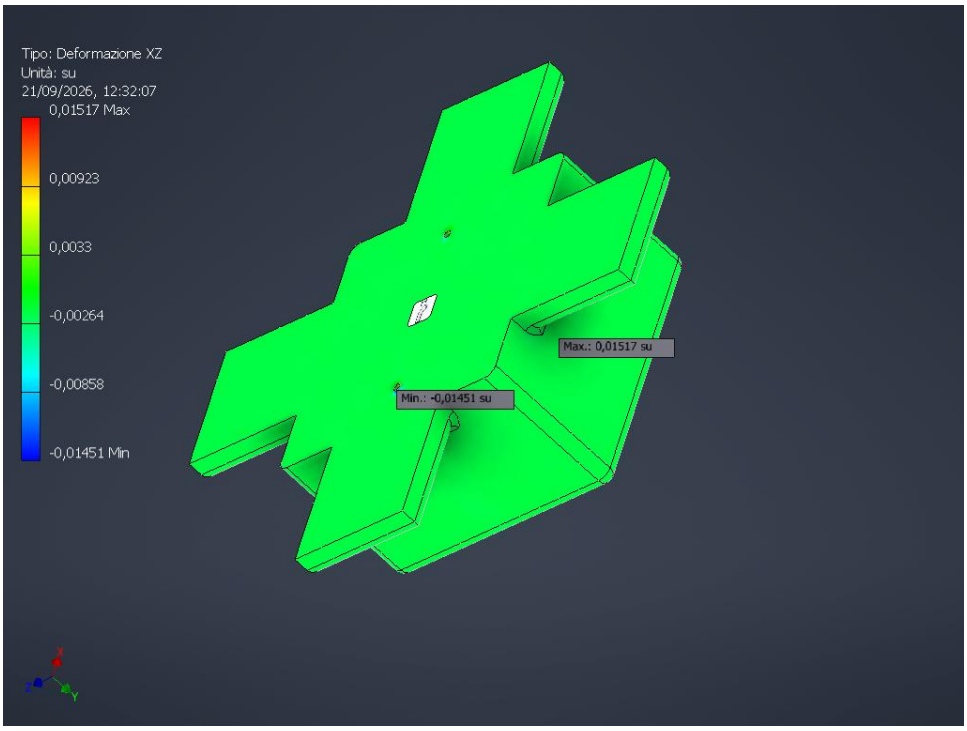
Deformazione XY



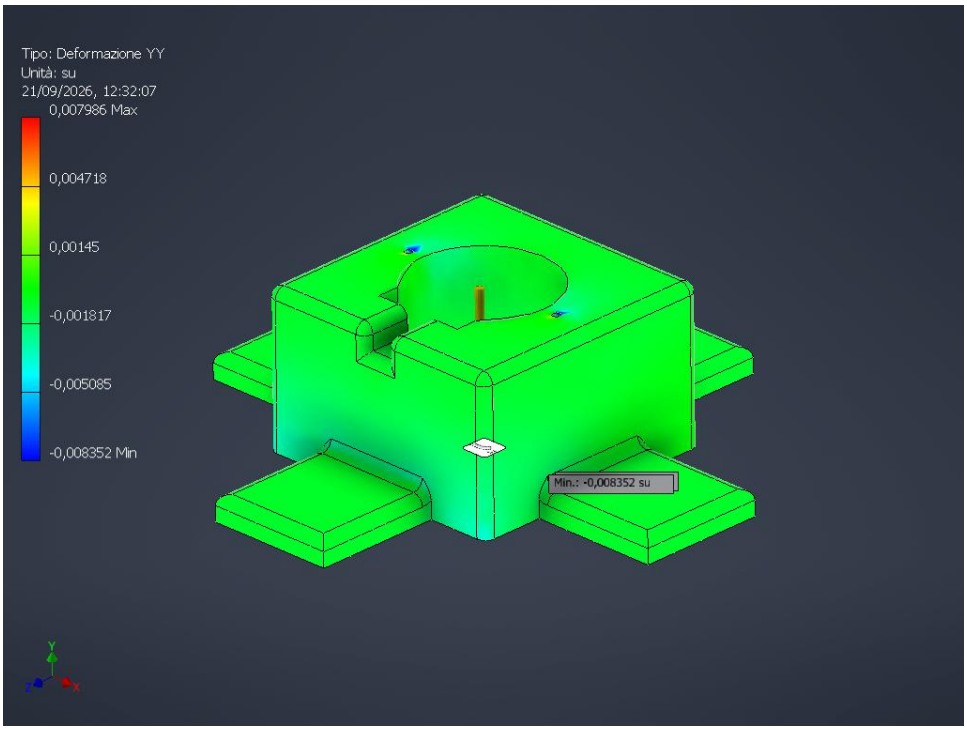


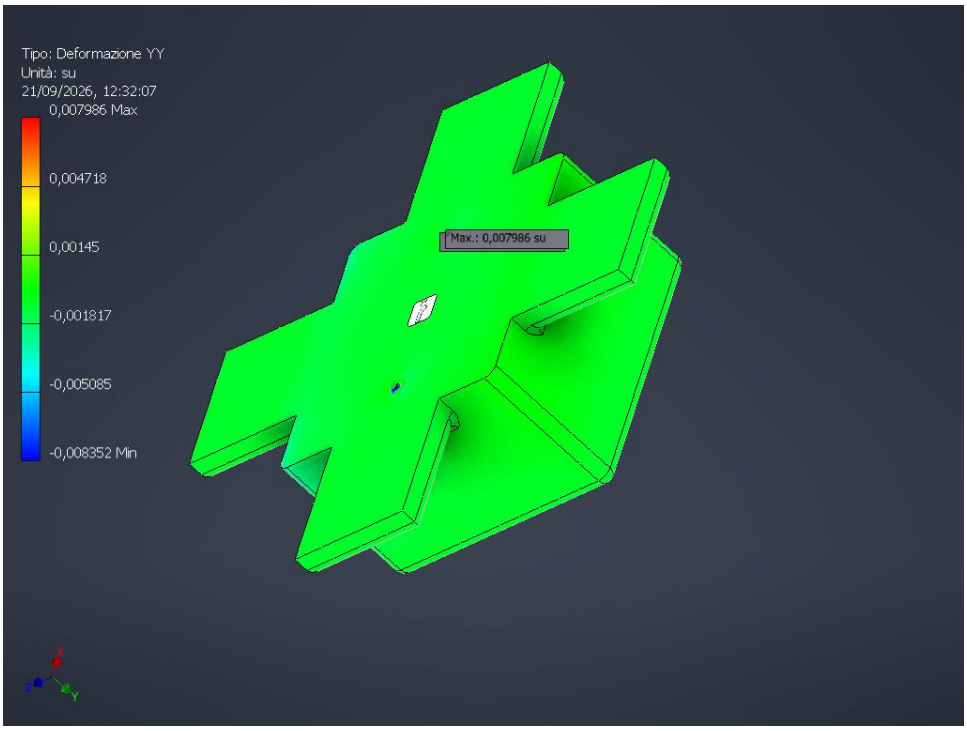
Deformazione XZ



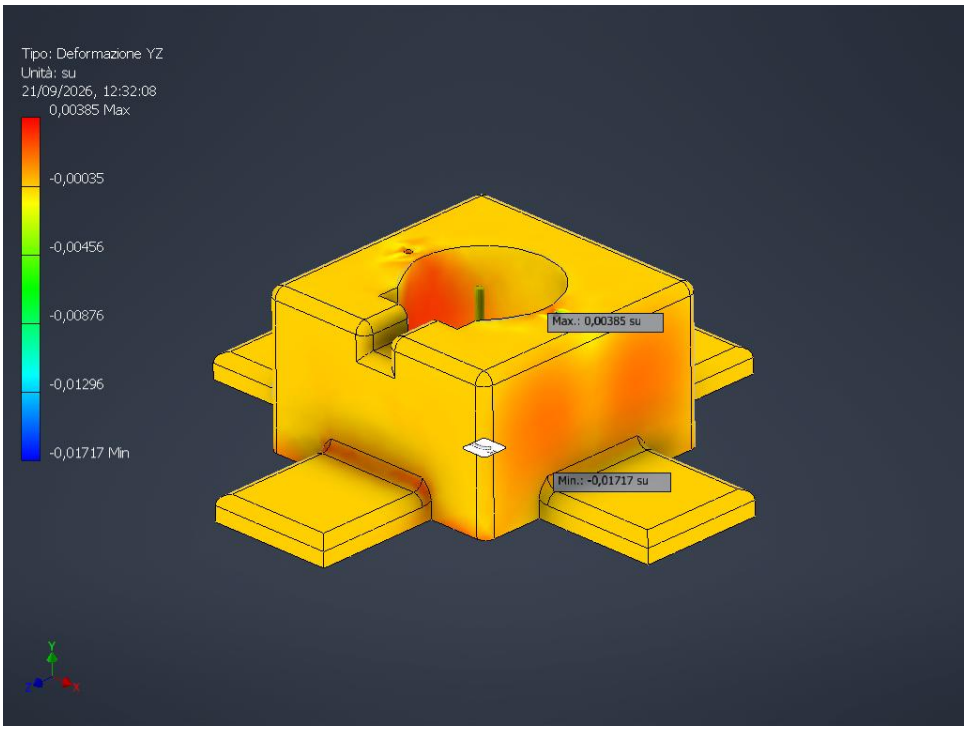


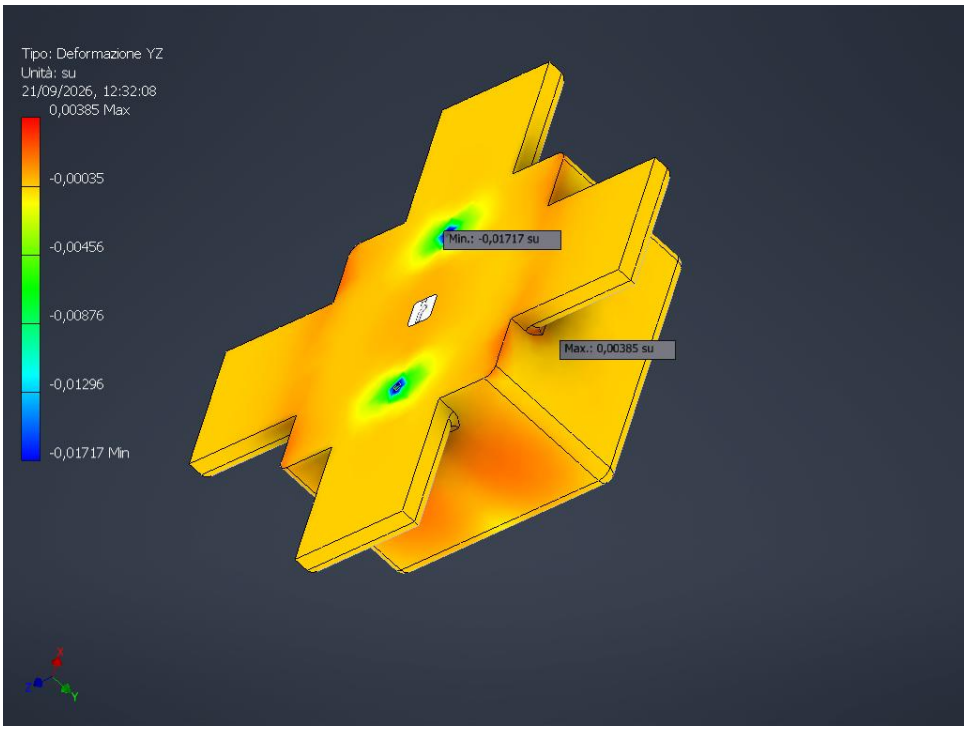
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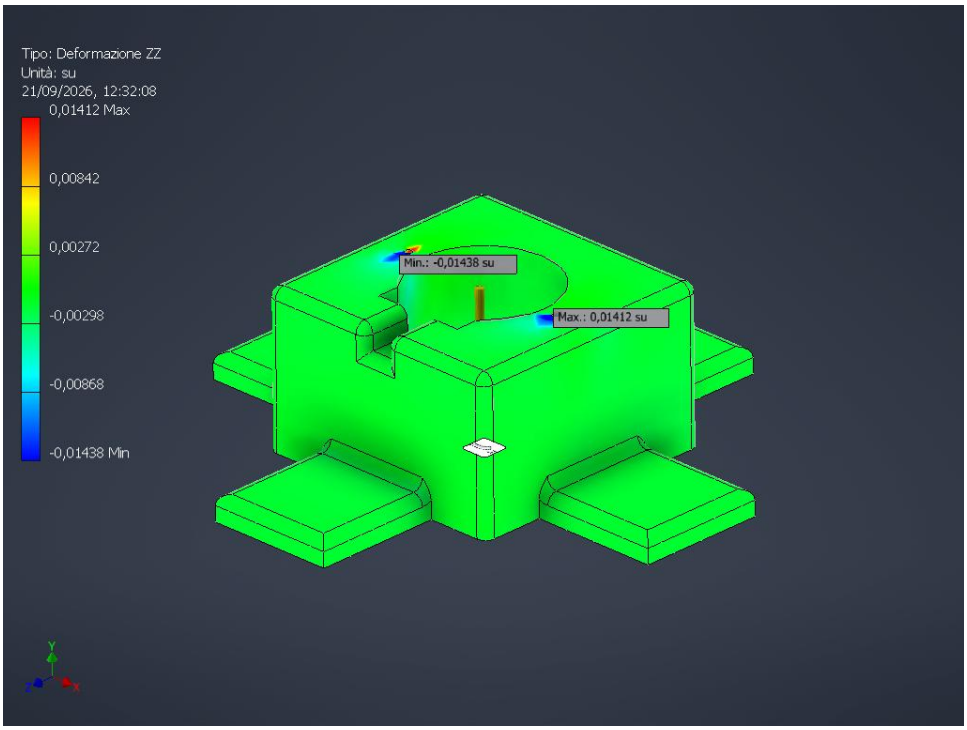


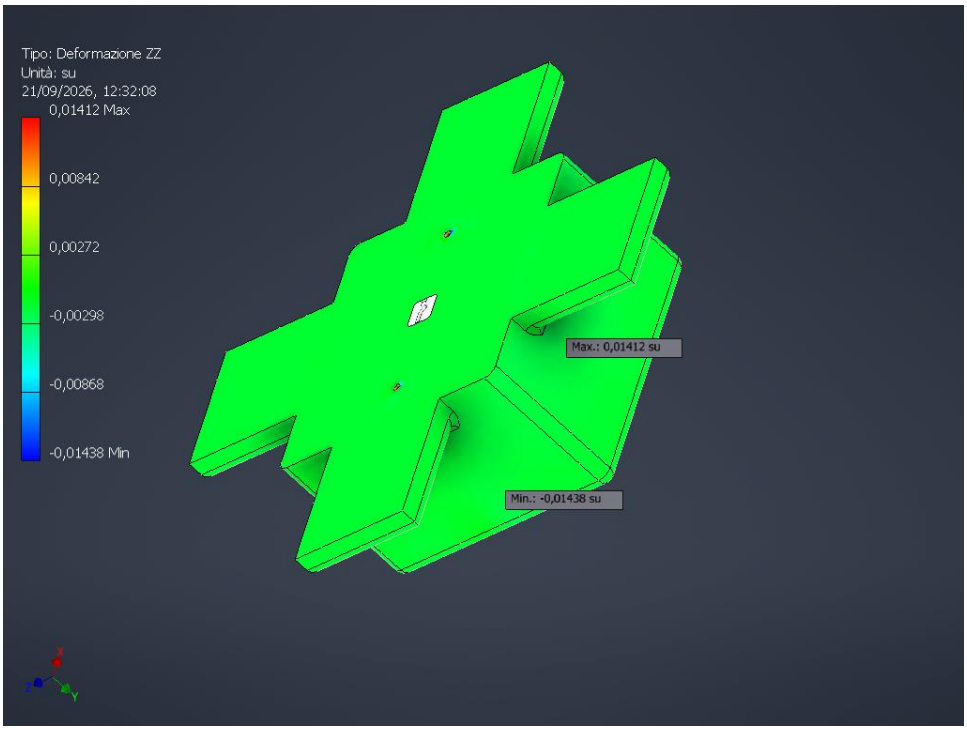
Deformazione YZ





Deformazione ZZ





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